

Layers of immunity: Deconstructing the *Drosophila* effector response

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This work provides one of the first **important** attempts to look at *Drosophila* immune responses against bacterial, viral, and fungal pathogens in a way that combines the roles of four major arms in immunity (Imd signaling, Toll signaling, phagocytosis, and melanization) rather than studying them separately. The findings are **compelling** and the tools provided can be used as they are, or built upon, in various contexts.

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Abstract

The host innate immune response relies on the cooperation of multiple defence modules. In insects and other arthropods, which have only innate immune mechanisms, four main immune-specific modules have been described in the defence against microbial invaders: the Toll pathway, the Imd pathway, the melanization response, and phagocytosis by plasmatocytes. Our present understanding of their relative importance remains fragmented as their contribution to host defence has never been simultaneously assessed across a large panel of pathogens. Here, we have taken advantage of newly-described immune mutants in a controlled genetic background to systematically delete these four immune modules individually, in pairs, or even all four simultaneously. Surprisingly, immune deficient flies simultaneously deficient in all four immune modules are viable, homozygous fertile, and display no overt morphological defects, suggesting these immune mechanisms are not strictly required for organismal development. With this new set of lines, we assessed the contribution of each module individually and collectively to host defense against five viruses, three fungi, eight Gram-positive bacteria, and eight Gram-negative bacteria. Our findings show that these four modules largely function independently and additively in host defense, although synergistic effects can occur for select pairs of modules. Our study confirmed the importance of the Imd pathway against Gram-negative bacteria and the Toll pathway against Gram-positive bacteria and fungi, largely via the induction of effectors such as antimicrobial peptides (AMPs) and Bomanins (Bom), but also reveals the important role of melanization against viruses, and the contribution of phagocytosis against various germs. Additionally, by examining microbial load kinetics in different mutants, we provide insight into how these modules contribute to tolerance or resistance against specific microbes. Our study provides insights into the architecture of the *Drosophila* immune system, revealing differential requirements of immune modules according to each pathogen. The set of immune deficient

lines provided here offer tools to better assess the role of these immune modules in host defense.

Introduction

The immune system is made up of a network of tissues and circulating cells whose primary function is to prevent and limit pathogenic infections (Danilova, 2006 [↗](#); Pradeu et al., 2024 [↗](#); Schmid-Hempel, 2021 [↗](#)). Although immunity has been studied in many animal species, it has been extensively characterized in only a subset of them such as the nematode *Caenorhabditis elegans*, the fruit fly *Drosophila melanogaster*, zebrafish, rodents, and humans. The traditional understanding of the immune system pictures it as a specialized group of cells and tissues, but this has gradually changed over recent years to a more complex picture of highly redundant multilevel defense modules. We have learned a great deal in the last few decades on the molecular mechanisms regulating each of these immune modules from recognition and signaling to effector action (ex. phagocytosis, antimicrobial peptides) (Pradeu et al., 2024 [↗](#)). However, an immune response encompasses the concomitant activation of several modules and there has been little investigation of how each module collectively contributes to host defense. Indeed, the immune research field has traditionally focused on one module at a time. This article aims to fill this gap by collectively analyzing the specific contribution of the main immune modules involved in the *Drosophila* systemic response.

The *Drosophila* systemic immune response consists of a set of humoral and cellular reactions in the hemolymph (insect blood) when flies are systemically infected by bacteria, viruses, or fungi (Liegeois and Ferrandon, 2022 [↗](#); Westlake et al., 2024 [↗](#)). A first facet of the systemic immune response is the production of host-defense effectors by the fat body and hemocytes that are secreted into the hemolymph. Among the best-characterized effectors are the antimicrobial peptides (AMPs), whose expression is induced to incredibly high levels upon infection in a race to control invading microbes (Hanson et al., 2019 [↗](#); Imler and Bulet, 2005 [↗](#)). AMPs are small cationic peptides that exhibit antimicrobial activity that directly participate in microbial killing. Beyond AMPs, many other proteins and host defense peptides are secreted into the hemolymph, creating a hostile environment for invading pathogens. The systemic antimicrobial response is predominantly regulated at the transcriptional level by the Toll and Imd pathways (De Gregorio et al., 2002 [↗](#); Lemaitre et al., 1996 [↗](#)). In *Drosophila*, the Toll pathway is activated by cell wall components (fungal glucans and Lysine-type peptidoglycan), as well as microbial proteases (Gottar et al., 2006 [↗](#); Issa et al., 2018 [↗](#); Leulier et al., 2003 [↗](#); Vaz et al., 2019 [↗](#)). Toll signaling culminates in the activation of two NF- κ B transcription factors, Dif and Dorsal, which regulate a large set of immune genes (Ip, 1993 [↗](#); Lemaitre et al., 1996 [↗](#)). These encode small peptides such as the antifungal peptide Drosomycin, the Toll-regulated Bomanin peptides, and many other proteins (e.g. serine proteases, serpins, lipases) (Clemmons et al., 2015 [↗](#); De Gregorio et al., 2002 [↗](#); Fehlbauer et al., 1994 [↗](#); Ligoxygakis et al., 2003 [↗](#)). Flies lacking the Toll pathway are viable but highly susceptible to infection by Gram-positive bacteria and fungi (Buchon et al., 2009 [↗](#); Lemaitre et al., 1996 [↗](#); Rutschmann et al., 2002 [↗](#)). The Imd pathway is activated by DAP-type peptidoglycan produced by Gram-negative bacteria and a subset of Gram-positive bacteria (e.g. *Bacillus*) (Kaneko et al., 2004 [↗](#); Lemaitre et al., 1995 [↗](#); Leulier et al., 2003 [↗](#)). The binding of peptidoglycan to receptors of the Peptidoglycan Recognition Protein family (PGRP-SD, PGRP-LC, PGRP-LE) initiates an intracellular signaling cascade, which ultimately activates the NF- κ B factor Relish (Choe et al., 2005 [↗](#); Iatsenko et al., 2016 [↗](#); Kaneko et al., 2006 [↗](#); Kaneko et al., 2006 [↗](#); Westlake et al., 2024 [↗](#)). The Imd pathway regulates the expression of many genes encoding antibacterial peptides, and serine proteases (De Gregorio et al., 2002 [↗](#); Lemaitre et al., 1997 [↗](#)). Imd deficient flies are viable but susceptible to Gram-negative bacterial infection.

In addition to humoral pathways, melanization, and phagocytosis are two complementary mechanisms that contribute to host survival upon systemic infection. Melanization is an arthropod-specific immune response resulting in the rapid deposition of the black pigment melanin at wound or infection sites and the concomitant production of microbicidal reactive oxygen species (Cerenius et al., 2008 [↗](#); Nappi et al., 2009 [↗](#); Westlake et al., 2024 [↗](#)). The melanization and Toll pathways are co-activated at the level of the serine proteases Hayan and Persephone, which in turn activate Spatzle-processing enzyme (SPE) of the Toll pathway and serine proteases critical to cleavage of prophenoloxidases (PPOs) (Dudzic et al., 2019 [↗](#); Nakano et al., 2023 [↗](#); Shan et al., 2023 [↗](#)), collectively referred to as the Toll-PO SP cascade (Westlake et al., 2024 [↗](#)). The melanization reaction itself relies on the oxidation of phenols, resulting in the polymerization of melanin. Two PPOs (PPO1 and PPO2) are the primary catalysts for the synthesis of melanin during the melanization response upon systemic infection. Flies deficient for both *PPO1* and *PPO2* are viable, but lack hemolymph phenoloxidase activity and exhibit susceptibility to certain Gram-positive bacteria and fungi (Binggeli et al., 2014 [↗](#); Dudzic et al., 2015 [↗](#)). In addition, a third PPO gene (*PPO3*) is specifically expressed by lamellocytes, specialized hemocytes that differentiate in larvae responding to and enveloping invading parasites (Dudzic et al., 2015 [↗](#)).

Phagocytosis by both sessile and circulating plasmatocytes (*Drosophila* equivalent of macrophages) is thought to provide a complementary and important host defense (Melcarne et al., 2019 [↗](#); Ulvila et al., 2011 [↗](#)). The use of hemocyte-deficient flies, via the targeted expression of a pro-apoptotic gene in plasmatocytes only, has shown that hemocytes contribute to survival upon systemic infection to certain bacterial species (Charroux and Royet, 2009 [↗](#); Defaye et al., 2009 [↗](#); Stephenson et al., 2022 [↗](#)). Phagocytosis of bacteria greatly relies on two transmembrane receptors, NimC1 and Eater (Bretscher et al., 2015 [↗](#); Kocks et al., 2005 [↗](#); Kurucz et al., 2007 [↗](#); Melcarne et al., 2019 [↗](#)). *NimC1*, *Eater* double mutant larvae are viable, have elevated hemocyte numbers at the larval stage, and their hemocytes can encapsulate and melanize macroparasites; however, their hemocytes are non-sessile and nearly totally phagocytosis deficient. *NimC1*, *Eater* adult flies have decreased hemocyte number and fail to perform phagocytosis, providing a good tool to assess the role of cellular response (Melcarne et al., 2019 [↗](#); Melcarne, 2020 [↗](#)).

We have circumstantial evidence that these four immune modules are critical for defense against certain pathogens. However, how, when, and to what extent each of these four immune modules contribute to host defense against specific pathogens is fragmented as they have never been simultaneously assessed. Moreover, the extent to which these findings can be generalized to broad classes of pathogens remains unclear.

In this article, we have used a specific set of mutations principally affecting phagocytosis (*NimC1*, *Eater*), melanization (*PPO1*, *PPO2*), the Toll pathway (*spz*), and the Imd pathway (*Rel*) in a controlled genetic background to address their role in host defense. We then recombined these and other mutations to generate double and even quadruple mutants of the four immune modules. This set of tools allowed us to compare the survival of flies deficient in these responses against a broad panel of viruses, fungi, Gram-positive bacteria, and Gram-negative bacteria, to reveal the relative contribution of each immune mechanism to resistance against these pathogens.

Material and methods

Insect stocks

Precise details of *Drosophila* stocks used in this study are provided in **Table 1** [↗](#). To minimize the influence of genetic background for the mutations used in this study, mutations were partially isogenized into the *DrosDel iso w¹¹¹⁸* genetic background (BDSC #5905) as described by Ferreira et al. (Ferreira et al., 2014 [↗](#); Ryder et al., 2004 [↗](#)). Isogenized lines were generated and used for

every mutant except the double mutant Toll and Imd (Rel^{E20} , spz^{rm7}) for which homozygous isogenized flies were not viable in our hands. Double mutants for $NimC1^1$; $Eater^1$ were poorly viable when isogenized in the *DrosDel iso w¹¹¹⁸* background, and so we used both isogenic and non-isogenic lines over the course of our study to facilitate investigation. Furthermore, the $NimC1^1$; spz^{rm7} , $Eater^1$ triple mutants were homozygous lethal isogenized or not, and so could not be included. In addition, the line $\Delta AMP14$ bears a knockout of 14 different AMP genes, which includes the AMP families Drosomycin, Metchnikowin, Cecropin, Defensin, Drosocin, Attacin, and Dipteracin, described previously (Carboni et al., 2022; Hanson et al., 2019). The Bom^{A55C} mutation was characterized in Clemmons et al. (Clemmons et al., 2015) and isogenic flies generated in Hanson et al. (Hanson et al., 2019). For larval experiments, use of GFP-labelled CyO or TM6,Tb balancers were used to enable picking of homozygous mutant larvae.

Drosophila lines	Reference	Name	Deficient pathway			
			IMD	Toll	Phagocytosis	Melanization
<i>iso</i> ; <i>iso</i> ; Rel^{E20}	Hedengren et al., 1999	ΔIMD	-	+	+	+
<i>iso</i> ; <i>iso</i> ; spz^{rm7}	Lemaitre et al., 1996	$\Delta TOLL$	+	-	+	+
<i>iso</i> ; $NimC1^1$; $Eater^1$	Melcane et al., 2019	$\Delta Phag$	+	+	-	+
<i>iso</i> ; $PPO1\Delta$, $PPO2\Delta$; <i>iso</i>	Bingelli et al., 2014	ΔMel	+	+	+	-
+ ; + ; Rel^{E20} , spz^{rm7}	this paper	ΔIMD , $\Delta TOLL$	-	-	+	+
<i>iso</i> ; $PPO1\Delta$, $PPO2\Delta$; $NimC1^1$, $Eater^1$	this paper	$\Delta Phag$, ΔMel	+	+	-	-
<i>iso</i> ; $NimC1^1$; $Eater^1$, Rel^{E20}	this paper	ΔIMD , $\Delta Phag$	-	+	-	+
<i>iso</i> ; $PPO1\Delta$ $PPO2\Delta$; Rel^{E20}	this paper	ΔIMD , ΔMel	-	+	+	-
<i>iso</i> ; $PPO1\Delta$ $PPO2\Delta$; spz^{rm7}	this paper	$\Delta TOLL$, ΔMel^{PPS}	+	-	+	-
<i>iso Hay-psh^{Def}</i> ; <i>iso</i> ; <i>iso</i>	Dudzic et al., 2019	$\Delta TOLL$, ΔMel^{HP}	+	-	+	-
<i>iso Hay-psh^{Def}</i> ; $NimC1^1$; $Eater^1$ Rel^{E20}	this paper	ΔTPM	-	-	-	-

Table 1

List of Mutants used in this study.

“-” indicates the deleted immune module.

Of note, the strains used in this study differ in their presence/absence of the *white*⁺ gene, present in the $PPO1^A$, $NimC1^1$ and *eater*¹ mutations. In addition to its well established function in eye pigmentation, the *white* gene can also impact host neurology and intestinal stem cell proliferation (Ferreiro et al., 2017; Sasaki et al., 2021). We did not observe any obvious correlations between *white*⁺ gene status and susceptibilities in this study. Moreover, in a previous study looking at the cumulative effects of AMP mutations on lifespan, *white* gene status and fluorescent markers did not readily explain differences in longevity (Hanson and Lemaitre, 2023). We therefore believe that the extreme immune susceptibility we have created through deficiencies for pathways regulating hundreds of genes, or major immune modules, overwhelms the potential effects of *white*⁺ and other transgenic markers. For additional information on which stocks bear which markers, see discussion in Supplementary file 1.

Microorganisms culture

Microbe strain information, microorganism classifications, and culture conditions are listed in Table S1. Bacterial cultures and *C. albicans* were grown overnight shaking at 200rpm. *Aspergillus fumigatus* fungus was grown on Malt Agar at room temperature until sporulation. The entomopathogenic fungus *Beauveria bassiana* strain R444 was provided by Andermatt AG as spore preparations (BB-PROTEC) which were directly used in natural infections or dissolved in PBS for septic injuries. Viruses DCV and FHV were kindly provided by Carla Saleh (Pasteur Institute, Paris),

produced in S2 cells. The supernatant of S2 cells was titrated before being used for fly infections. Viruses SINV, DXV and IIV-6 were kindly provided by Ronald van Rij (Radboud University Medical Center, Nijmegen).

Heat-killed microbes were prepared by two repeats of boiling at 95°C for 30 minutes then freezing at -20°C for 30 minutes, before storage long term at -20°C. Microbe preparations were streaked onto agar plates to check for full efficiency of heat-killing before use in experiments.

Gene expression

Flies were inoculated by pricking in the junction of thoracic pleura with a needle dipped in a mixed pellet containing a 1:1 mixture of OD600 = 200 heat-killed *E. coli* and *M. luteus* (final OD600 = 100 for each), and frozen at -20°C 6h, 12h and 24h post-infection. Total RNA was then extracted from pooled samples of five flies using TRIzol reagent per manufacturer's protocol and resuspended in MilliQ dH₂O. Reverse transcription was performed using PrimeScript RT (Takara) with random hexamer and oligo dT primers. Quantitative PCR was performed on a LightCycler 480 (Roche) in 96-well plates using Applied Biosystems PowerUP SYBR green Master Mix (Applied Biosystems). Data points represent pooled samples from three replicate experiments. Error bars represent one standard deviation from the mean. Statistical analyses were performed using one-way ANOVA with Holms-Sidak multiple test correction. Primers used in this study were:

Drs-F 5'- CGTGAGAACCTTTTCCAATATGAT-3'

Drs-R 5'- TCCCAGGACCACCAGCAT-3'

DptA-F: 5'-GCTGCGCAATCGCTTCTACT-3'

DptA-R: 5'-TGGTGGAGTGGGCTTCATG-3'

RPL32-F: 5'-GACGCTTCAAGGGACAGTATCTG-3'

RPL32-R: 5'-AAACGCGTTCTGCATGAG-3'

Ex vivo larval hemocyte phagocytosis assays

Ex vivo phagocytosis assays were performed using a mix of equal volumes of *E. coli* and *S. aureus* AlexaFluorTM488 BioParticlesTM (Invitrogen), following manufacturer's instructions; and see (Melcarne et al., 2019 [\[4\]](#)). Five L3 wandering larvae from our newly generated mutant lines, or carrying the *Hml-Gal4*, *UAS-GFP* hemocytes marker as a control, were bled into 150µL of Schneider's insect medium (Sigma-Aldrich) containing 1µM phenylthiourea (PTU; Sigma-Aldrich). The hemocyte suspension was then transferred to 1.5 mL low binding tubes (Eppendorf, SigmaAldrich), and AlexaFluorTM488 bacteria BioParticlesTM were added. The samples were vortexed, incubated at room temperature for 2 hours to allow phagocytosis and then placed on ice to stop the reaction. The fluorescence of extracellular particles was quenched by adding 0.4% trypan blue (Sigma-Aldrich) diluted to 1/3 concentration. Phagocytosis was quantified using a flow cytometer (BD Accuri C6) to measure the fraction of cells phagocytosing, and their fluorescence

intensity. Isogenic wild-type iso w^{1118} larvae and *Hml-Gal4, UAS-GFP* larvae with or without bacterial particles were used to define the gates for hemocyte counting and the thresholds for phagocytosed particle emission. The phagocytic index was calculated as follows:

$$\text{Fraction of haemocytes phagocytosing}(f) = \frac{[\text{number of haemocytes in fluorescence positive gate}]}{[\text{total number of haemocytes}]}$$

$$\text{Phagocytic index}(PI) = [\text{Mean fluorescence intensity of haemocytes in fluorescence positive gate}] \times f$$

Finally, due to experimenter and experiment batch differences in total hemocytes collected and knock-on effects to calculating phagocytic index, phagocytic index was normalised with the wild-type as a reference set to 100% within experiment batch.

Wounding experiment

Clean injury was performed with a needle sterilized with an EtOH and PBS wash. For imaging of the melanization reaction upon pricking, the thorax of 3-8 days old flies was pricked using a sterile needle (diameter ~0.1mm). Pictures were taken 24h post-pricking and categorized into normal, weak, or no melanization blackening seen at the injury site per Dudzic et al. (Dudzic et al., 2019 [DOI](#)).

Systemic infections and survival

Systemic infections were performed by pricking 3- to 5-day-old adult males in the thorax with a 0.1mm- thick needle dipped into a concentrated pellet of bacteria, yeast, or fungal spores. For natural infections with *A. fumigatus*, flies were anesthetized and then shaken on a sporulating plate of fungi for 30 seconds. For *B. bassiana*, flies were shaken for 30s in a tube containing an excess of commercial spores sufficient to coat all flies uniformly, of which the flies remove the vast majority later through grooming. At least two replicate survival experiments were performed for each infection, though in specific cases some genotypes may have been included in only one experiment. 20-35 flies were included per vial when possible, kept on cornmeal fly media containing (per 1L): 6.2g agar, 58.8g cornmeal, 58.8g yeast, 60mL grape fruit juice, 4.83mL propionic acid, and 26.5mL 100g/L moldex in pure ethanol. Survivals were scored daily, and flies were flipped to new food vials 3 times per week. Due to challenges working with genotypes of different viability, we could not include all genotypes in all experimental replicates. Moreover, comparisons across genotypes required multiple levels of consideration. Given the complexity of genotype-vs-reference-by-pathogen comparisons in our study, needing to compare infections both to clean injury within genotype, and to wild-type or $\Delta ITPM$ across treatments, we chose to focus on major differences apparent in summary statistics, highlighting survival differences only when they were: **i)** consistent across experimental replicates; **ii)** of a consistent logic across comparable genotypes; for instance, compound mutants containing ΔMel (e.g. ΔIMD , ΔMel) should be as or more susceptible to infections as ΔMel alone if melanization is truly critical for defence; and **iii)** of a mean lifespan difference ≥ 1.0 days after accounting for comparisons with unchallenged or clean-injury data. Total experiments (N exp) and total flies per genotype are reported to provide summary statistics per pathogen-by-genotype interaction. The mean lifespan reported in survival summary data has a maximum of 7 or 10 days depending on experimental conditions. Kaplan-Meier survival curves for all experiments are provided in the main text or supplementary information.

Quantification of microbial load for growth kinetics

3-8 days old flies were infected with the indicated microbe and concentration per OD600 as described in Table S1, and allowed to recover. At the indicated time post-infection, flies were anesthetized using CO₂ and surface sterilized by washing them in 70% ethanol. Ethanol was removed, and then flies were homogenized using a Precellys bead beater at 6500 rpm for 30 s in

LB broth (*Pr. rettgeri*), BHI (*S. aureus*), or YPG (*C. albicans*) with 500 μ L as pools of 5 flies. These homogenates were serially diluted and 100 μ L was plated on LB, BHI or YPG agar. Plates were incubated overnight, and colony-forming units (CFUs) were counted using a Interscience Scan 500 plate scanning colony counter, and validated independently by manual counts for a subset of plates to ensure accuracy. Statistical differences were tested using Brown-Forsythe and Welch ANOVA tests with Dunnett's multiple test correction, which consider unequal variance.

Results

A set of immune-deficient lines in the iso Drosdel background

Mutations affecting only one of the four main modules of the *Drosophila* systemic immune response have been described. These include *Relish*^{E20} (*Rel*^{E20}, deficient for the Imd pathway, referred to as Δ IMD), *spatzle*^{rm7} (*spz*^{rm7}, deficient for the Toll pathway, referred to as Δ TOLL), *PPO1* ^{Δ} , *PPO2* ^{Δ} (no melanization, referred to as Δ Mel), and *NimC1*¹; *eater*¹ (reduced phagocytosis, referred to as Δ Phag). In addition, to blocking phagocytosis, Δ Phag flies have a defect in hemocyte sessility, an increased total number of larval hemocytes, and a decreased number of hemocytes at the adult stage (Bretscher et al., 2015; Melcarne et al., 2019; Melcarne, 2020). We have preferred to use *NimC1*¹; *eater*¹ to study the contribution of hemocytes rather than other approaches with important indirect effects such as the pre-injection of latex beads to saturate phagocytes (Elrod-Erickson et al., 2000), or expression of a pro-apoptotic gene in plasmotocytes (Charroux and Royet, 2009; Defaye et al., 2009). All the mutations in this study were partly isogenized for seven or more generations into the DrosDel iso w¹¹¹⁸ genetic background (Ferreira et al., 2014; Ryder et al., 2004). We then generated by recombination six out of the seven combinations of flies simultaneously lacking two immune modules with the exception of Δ TOLL, Δ Phag as *NimC1*¹; *eater*¹, *spz*^{rm7} flies were not viable (see fly strains in Table 1). Of note, in our hands recombinant *Rel*^{E20}, *spz*^{rm7} flies were not viable in the DrosDel isogenic background forcing us to use non-iso flies to assess the host defense of Δ IMD, Δ TOLL flies. Over the course of our study, isogenic *NimC1*¹; *eater*¹ flies also displayed viability issues, and specific experiments used non-isogenic flies to improve confidence in phenotypes; these experiments are noted in figure captions. We used two approaches to simultaneously delete the Toll and Melanization modules: **i)** *PPO1* ^{Δ} , *PPO2* ^{Δ} ; *spz*^{rm7} flies (referred to as Δ TOLL, Δ Mel^{PPS}) and **ii)** flies carrying a small viable deletion *Hayan-psh*^{Def} (referred to as Δ TOLL, Δ Mel^{HP}) that removes two clustered serine protease genes, *Hayan*, and *persephone*, which blocks the TOLL-PO SP cascade that regulates both humoral-Toll and melanization (Dudzic et al., 2019; Westlake et al., 2024). This allowed us to delete the Toll and melanization pathways using only one deletion event on the X chromosome. Excitingly, we succeeded in generating a fly line deficient for all four immune modules with the genotype: *Hayan-psh*^{Def}; *NimC1*¹; *eater*¹, *Rel*^{E20} hereafter named Δ ITPM (Δ IMD, Δ TOLL, Δ Phag, Δ Mel). These flies were homozygous viable with very poor viability, and homozygous fertile, despite major deficiencies in these four important systemic immune mechanisms. These flies allow us to monitor survival kinetics and pathogen growth in the near-total absence of an immune system. Finally, we added two iso fly lines in our analysis that delete key host defense peptides downstream of the Toll and Imd pathways: **i)** *Bom*^{A55C}, a short deletion removing ten Bomanins at position 55C, which causes a major susceptibility to infection by Gram-positive bacteria and fungi (Clemmons et al., 2015), and **ii)** Δ AMP14, a compound fly line of eight mutations removing 14 AMP genes (4 Cecropins, 4 Attacins, 2 Dipterocins, Drosocin, Defensin, Drosomycin, and Metchnikowin), which causes a major susceptibility to infection by Gram-negative bacteria (Carboni et al., 2022; Hanson et al., 2019). This set of immune-deficient fly lines in a defined genetic background provides a unique tool kit to address the function of immune modules individually or collectively.

Imd, Toll, Phagocytosis, and Melanization modules function largely independently

The Imd, Toll, Phagocytosis, and Melanization modules are interconnected (Westlake et al., 2024 [↗](#)). For instance, the Toll and Imd pathways both regulate a subset of genes involved in melanization (De Gregorio et al., 2002 [↗](#); Ligoxygakis et al., 2002 [↗](#)). Moreover, one module can indirectly affect another. For instance, a lack of melanization could increase the bacterial load after infection, resulting in higher activation of the Toll and Imd pathways (Binggeli et al., 2014 [↗](#)). In the first step, we analyzed the activation of each of the four modules in single- and multiple-module deficient flies using classical readouts of these modules. This also allowed monitoring of the possible effects of deleting one module on the activity of another. We analyzed the immune response in single module deficient flies by monitoring **i)** the expression of the antibacterial peptide gene *Diptericin* (*DptA* - a target of the Imd pathway), and **ii)** the antifungal peptide gene *Drosomycin* (*Drs* - a target of the Toll pathway) upon systemic infection with a mixture of heat-killed *Escherichia coli* and *Micrococcus luteus*, **iii)** ex vivo phagocytosis of *E. coli* and *Staphylococcus aureus* coated bio-particles incubated with larval hemocytes, and **iv)** melanization at the injury site of adult flies. This analysis first confirmed that Δ IMD, Δ TOLL, Δ Phag, and Δ Mel flies are deficient for Imd, Toll, phagocytosis, and melanization respectively (**Figure 1A, B, C, D** [↗](#)). We note that the expression level of *DptA* was roughly similar to the wild-type in Δ TOLL and Δ Mel flies, however, Δ Phag flies show reduced *DptA* induction at 6h compared to WT (~50%) (**Figure 1A** [↗](#)). This still represents an immense induction of *DptA* after infection, but may suggest a lag or lesser overall Imd response in Δ Phag flies. *DptA* expression in Δ Phag was, however, comparable to wild-type at 12h and 24h. On the other hand, the *Drs* induction at 6h of Δ Mel flies was significantly higher than WT (~1.6x), as found previously using live *M. luteus* infection (Binggeli et al., 2014 [↗](#)) (**Figure 1B** [↗](#)). The phagocytic assay confirmed that the Δ Phag mutant was significantly impaired in its ability to phagocytose bacterial bio-particles. Furthermore, Δ IMD, and Δ Mel mutant larvae had similar levels of phagocytic index to the wild-type (**Figure 1C** [↗](#)), while Δ TOLL had a higher phagocytic index, if anything. As expected, there was no apparent melanization at the site of the injury in Δ Mel flies while melanization blackening was readily observed in the wild-type and in other single-module mutants (**Figure 1D** [↗](#)). Collectively, this analysis indicates that each of the four immune modules can be selectively blocked in the isogenic DrosDel background without interfering with the competence of the others. We sometimes noticed a modest difference of module activity in the absence of another (higher Toll activity in Δ Mel, lower IMD activity in Δ Phag and higher phagocytic activity in Δ TOLL), findings that require further exploration to be fully established.

We then repeated our analysis using our set of module compound mutants. Not surprisingly, all compound flies display the expected immune deficiencies validating the lines we constructed. This analysis confirms that Δ ITPM flies, although viable, are indeed fully immune deficient for the four host defense modules. We did not detect any striking synergistic or antagonistic effects in which deleting two modules has a vastly different impact compared to the additive effect of deleting the two modules individually. Notably, a small level of *Drs* expression is retained in Δ TOLL mutant flies, but was fully abolished in Δ IMD, Δ TOLL flies (**Figure 1A vs 1E** [↗](#)), confirming the *Drs* gene receives a small input from the Imd pathway (Leulier et al., 2000 [↗](#)). The Δ TOLL, Δ Mel^{PPS} and, Δ TOLL, Δ Mel^{HP} flies display *Drs* induction at 6h that is resolved to unchallenged WT levels by 12h, consistent with this induction coming from an early input from the IMD pathway in these backgrounds. Taken together, all the compound mutant flies behave largely as expected given the sum effects of their individual mutations, confirming that these modules largely function independently.

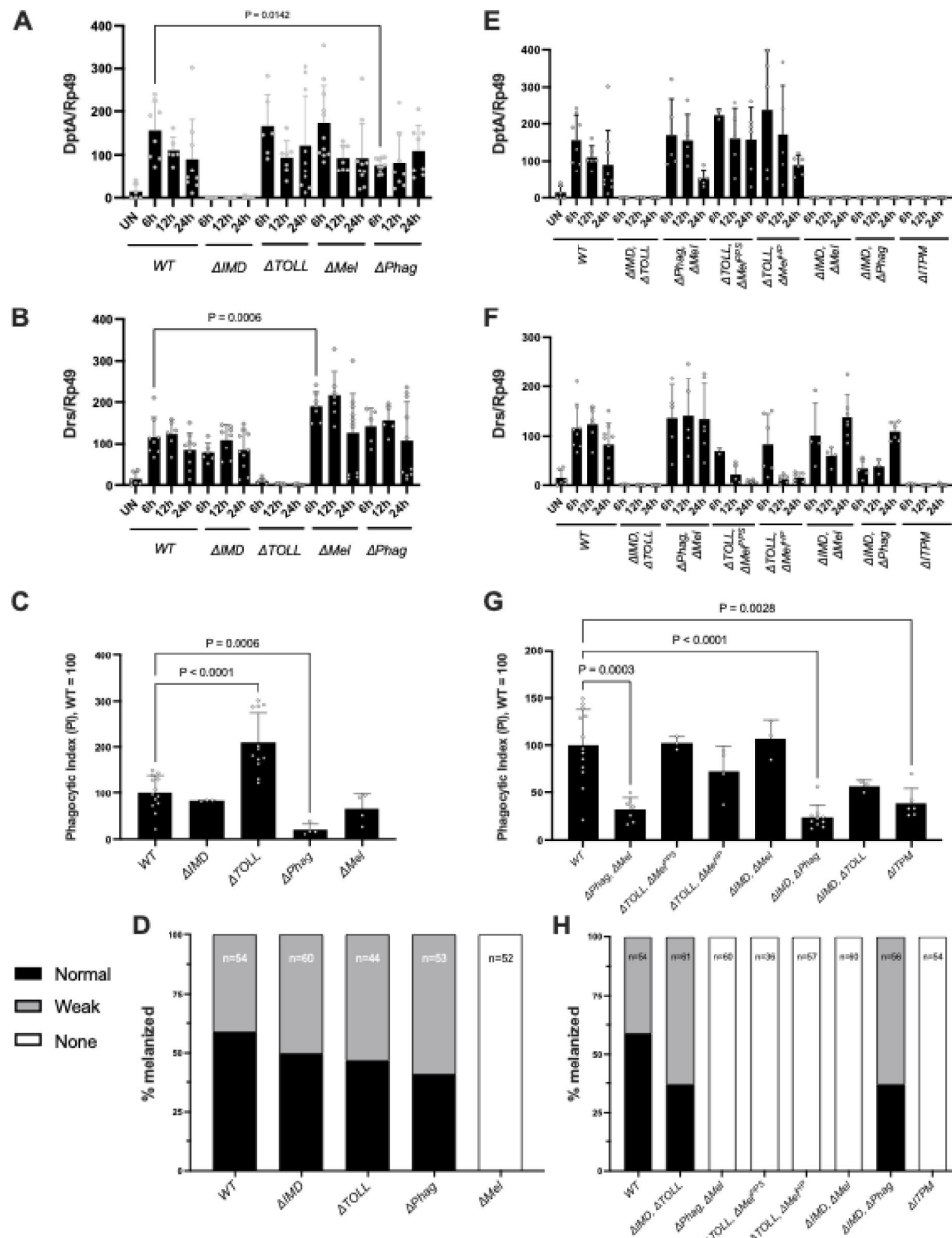


Figure 1

Each immune module functions largely independently from other immune modules.

A) Activation of the Imd pathway, as revealed by *DptA* expression following infection with a mixture of heat-killed *E. coli* and *M. luteus*, is broadly wild-type in single module mutants other than Imd. In the case of Δ Phag flies, we observed a lesser induction at 6h compared to wild-type. **B**) Activation of the Toll pathway, as revealed by the expression of *Drs* is broadly wild-type in single module mutants other than Δ TOLL flies. In the case of Δ Mel, we observed a slightly greater Toll pathway activity. **C**) The ability of plasmatocytes to phagocytose bacterial particles is not negatively affected in single module mutants except Δ Phag (also see [Figure 1supp1](#)). **D**) Cuticle blackening after clean injury is not impaired except in Δ Mel flies. **E**) Activation of the Imd pathway remains strongly inducible in compound mutants except when deficient for the Imd pathway. **F**) Activation of the Toll pathway remains strongly inducible in compound mutants except when flies were deficient of the Toll pathway. Note that *Drs* receives a minor input from Imd signaling (Leulier et al., 2000), explaining minor induction of *Drs* in Δ Toll flies at early time points. **G**) The ability of plasmatocytes to phagocytose bacterial particles is not significantly affected in compound mutants except those including Δ Phag. **H**) Cuticle blackening after clean injury is not impaired in compound mutants except in Δ Mel flies.

Both the Melanization and TOLL modules contribute to wounding and lifespan

Before assessing the resistance to infection, we monitored fly survival at 25°C and 29°C to determine if these mutations impact viability in the absence of challenge. At 25°C, no single mutants died significantly more than the wild-type in the absence of challenge within the ten-day window (**Figure 2A**). However, looking at compound mutants, even in the absence of challenge, each compound mutant other than ΔIMD , $\Delta TOLL$ suffered 25% cumulative mortality by 10 days post-eclosion, while $\Delta TOLL$, ΔMel^{PPS} and $\Delta ITPM$ flies suffered even greater cumulative mortality. Similar trends were observed at 29°C with slightly elevated mortality (**Figure 2supp1**). The observation that $\Delta TOLL$, ΔMel^{PPS} ($PPO1^1, PPO2^1, spz^{rm7}$) flies have shorter lifespan compared to $\Delta TOLL$, ΔMel^{HP} (*Hayan-psh^{Def}*) could be explained by residual Toll and/or PO activity in *Hayan-psh^{Def}* or additional indirect effects in $\Delta TOLL$, ΔMel^{PPS} flies. As such, the combined effect of Toll and Melanization deficiencies in unchallenged and clean injury survival is best assessed by considering signals from both genotypes. $\Delta ITPM$ flies (*Hayan-psh^{Def}*; *NimC1¹*; *Eater¹*, *Rel^{E20}*) have the lowest unchallenged survival rate of genotypes included in this study. Taken together, the Imd, Toll, Melanization, and Phagocytosis modules contribute little to lifespan maintenance individually, but are crucial to maintain lifespan collectively (**Figure 2B**).

Systemic infections were performed by septic injury (pricked with a contaminated needle). We therefore monitored the survival of single and double mutant flies after a clean injury to disentangle the effects of injury from those of infection. We found a cumulative mortality of ~50% for ΔMel flies by day 10 (**Figure 2C**), while other single module mutants displayed relatively little mortality over the 10-day window; ΔIMD had notable late-onset mortality in a minority of experimental replicates, which may be due to stochastic dysbiosis effects during aging (Hanson and Lemaître, 2023; Marra et al., 2021). Double mutant ΔIMD , $\Delta TOLL$ and ΔIMD , $\Delta Phag$ flies retained a high survival rate. Consistent with an importance of melanization in the injury response (Binggeli et al., 2014), all the compound mutants affected in melanization were susceptible to clean injury, with similar mortality seen in $\Delta Phag$, ΔMel , ΔIMD , ΔMel , $\Delta TOLL$, ΔMel^{PPS} , $\Delta TOLL$, ΔMel^{HP} , and $\Delta ITPM$ flies (**Figure 2D**). Thus, while these genotypes differ in their base lifespan, their viability upon injury is remarkably similar. These results emphasize an interconnectedness of the melanization response with each other immune module for maintaining health after injury.

Similar results were obtained for both unchallenged and clean injury lifespans at 29°C. Of note, $\Delta TOLL$, ΔMel^{PPS} suffered significantly greater mortality in unchallenged conditions than other genotypes, even including $\Delta ITPM$ flies (**Figure 2supp1**). Other compound mutant genotypes displayed a continuum of mortality in unchallenged conditions between $\Delta ITPM$ (~50% cumulative mortality) and $\Delta TOLL$, ΔMel^{HP} (~15% cumulative mortality). Upon clean injury, results were broadly consistent with 25°C trends, with a more uniform mortality across all genotypes.

Collectively, our study highlights that all these immune modules contribute somewhat to the survival of flies upon clean injury, with a major role of the melanization response in maintaining lifespan and survival to injury. In the next steps of our study, we compared survival upon infection to both unchallenged and clean injured flies to distinguish the impact of microbial infection from the injury itself.

Multiple mechanisms contribute to resistance to microbial infection

To compare the contribution of each module to host defense, we performed infection experiments of wild-type and single- and multiple-module mutant flies following septic infection with five viruses, eight Gram-positive bacteria, eight Gram-negative bacteria, septic injury for two fungal

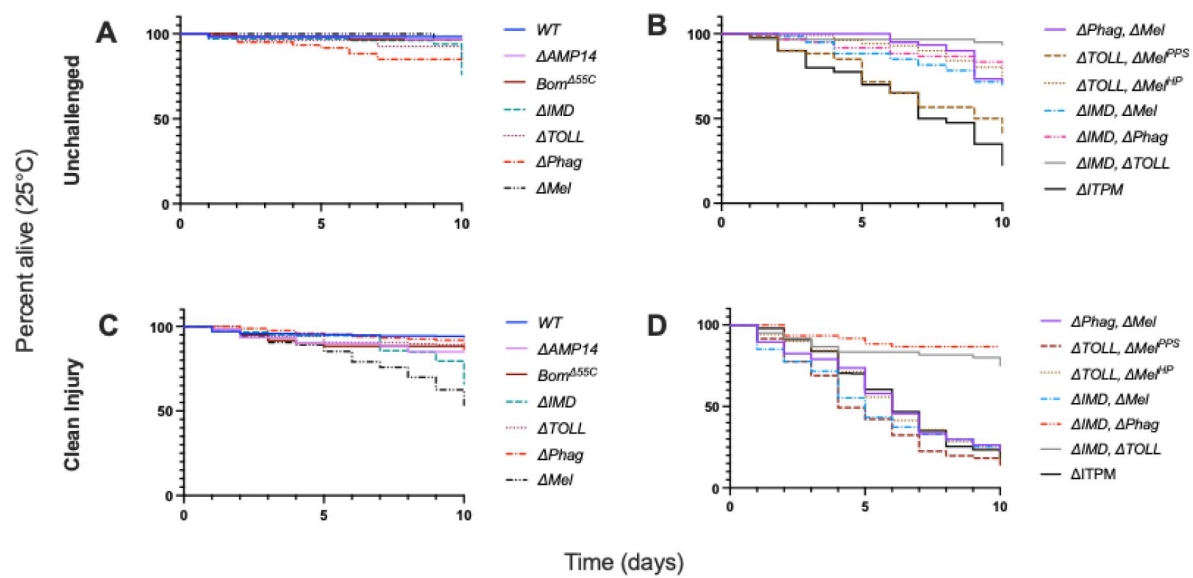


Figure 2

Lifespans of mutants used in this study in unchallenged flies (A-B) and upon clean injury (C-D) conditions at 25°C.

See [Figure 2supp1](#) for 29°C comparisons.

species, and natural infection (i.e. spore deposition on the cuticle) for two fungal species. Temperature, dose, and days monitored are shown in **Figure 3**, and see Table S1 for other key parameters. Survival curves are shown in Supplementary file 2 for each pathogen and results of these survivals are summarized in **Figure 3** (and Supplementary file 3) as mean lifespans given 7 or 10 days as the maximum lifespan per treatment, with number of tested flies indicated on the right and darker blue intensity as indicative of lower lifespan.

We performed infection experiments with the following viruses and microbial species: DCV, FHV, DXV, IIV-6, SINV, *Aspergillus fumigatus*, *Beauveria bassiana*, *Candida albicans*, *Mycobacterium marinum*, *Corynebacterium diphtheriae*, *Micrococcus luteus*, *Streptococcus pneumoniae*, *Enterococcus faecalis*, *Staphylococcus aureus*, *Listeria monocytogenes*, *Bacillus subtilis*, *Vibrio parahaemolyticus*, *Providencia rettgeri*, *Providencia burhodogranariae*, *Pectobacterium carotovorum* Ecc15, *Klebsiella pneumoniae*, *Enterobacter cloacae*, *Escherichia coli*, and *Salmonella enterica* ser. typhimurium. Collectively, this represents infections with 24 pathogens across 14 genotypes and two infection routes (336 genotype-by-pathogen interactions). When combined with comparisons to clean injury and unchallenged controls to validate if a lifespan difference is meaningful, and to wild-type or $\Delta ITPM$ treatments, this inflates to 1000+ interactions. We therefore focused our survival analysis on major effects. We report the core summary statistics and overt trends, and comment only where difference in mean lifespan between reference genotypes or control treatments was a minimum of 1 day.

Our study confirms that the Imd pathway plays a major role against all tested Gram-negative bacteria, as ΔIMD flies succumb more rapidly than wild-type flies to all of them. This pathway also contributes, albeit to a lesser extent, to survival against the Gram-positive bacteria *B. subtilis* and *L. monocytogenes*, which bear DAP-type peptidoglycan at their membrane that triggers Imd activation (Kaneko et al., 2004; Leulier et al., 2003). We were interested to see the impact of our ΔIMD flies in host defense to viruses as the specific mutation we used (*Rel^{E20}*) deletes a transcription factor common to both the canonical Imd pathway and the recently- appreciated cGLR-Sting-Relish pathway (Goto et al., 2018). We found that ΔIMD mutant flies were somewhat susceptible to the four viruses DCV, FHV, DXV, and IIV6, but not SINV, agreeing with previous studies showing a reduction in lifespan of 1-2 days for *Rel* mutant flies after viral infection (Costa et al., 2009; Goto et al., 2018; Sansone et al., 2015).

This study also confirms a prominent role of the Toll pathway in survival upon infection by all virulent Gram-positive bacteria and fungi upon septic injury. Less virulent microbes and infectious routes (such as *A. fumigatus* by natural infection) did not result in increased mortality in individual $\Delta TOLL$ mutant flies. A contribution of the Toll pathway to survival was also observed for some Gram-negative bacteria, including a minor contribution to survival upon *Pr. rettgeri* infection, and a major contribution after *V. parahaemolyticus* infection. In addition, we found a susceptibility of $\Delta TOLL$ flies to FHV, DXV, and IIV6, suggesting Toll-regulated effectors mediate defence against viral infection.

Use of $\Delta Phag$ mutants reveals a role of the cellular response in survival against the fungus *B. bassiana*. $\Delta Phag$ flies further show a minor susceptibility to *S. aureus*, and a subset of Gram-negative bacteria (*Pr. rettgeri*, Ecc15, and *E. coli*). Susceptibilities to Gram-negative bacteria could result from the observed lag in Imd pathway activation (**Figure 1A**); indeed, previous studies have shown a full *DptA* response is pivotal for defence against *Pr. rettgeri* (Hanson et al., 2023, 2019; Unckless et al., 2016), and lifespans are greatly impacted by subtle differences when using pathogens with intermediate levels of virulence (Duneau et al., 2017a). A role for Imd in explaining $\Delta Phag$ susceptibilities does not detract from the importance of the cellular response, but instead offers a mechanistic route to explore for these phenotypes.

Type	Species	Temp	Dose	Days	N exp	WT	$\Delta AMP14$	ΔBom	ΔIMD	$\Delta TOLL$	$\Delta Phag$	ΔMel	$\Delta Phag, \Delta Mel$	$\Delta TOLL, \Delta Mel^{PP3}$	$\Delta TOLL, \Delta Mel^{OP}$	$\Delta IMD, \Delta Mel$	$\Delta IMD, \Delta Phag$	$\Delta IMD, \Delta TOLL$	$\Delta JTPM$
Corrinos	Unchallenged	25°C	n/a	7	13	6.9	6.9	6.9	6.8	6.8	6.6	7.0	7.0	6.0	6.8	6.6	6.6	6.8	5.8
	Clean Injury	25°C	n/a	7	36	6.7	6.6	6.5	6.7	6.6	6.8	6.4	5.3	4.6	5.4	4.7	6.5	6.2	5.5
	Clean Injury	29°C	n/a	7	11	6.6	6.5	6.0	6.3	6.1	6.6	6.4	4.9	3.0	4.8	3.7	6.0	5.6	3.9
	Clean Injury	25°C	n/a	10	36	9.6	9.1	9.2	9.2	9.3	9.6	9.5	6.2	5.2	6.3	5.6	9.1	8.7	6.3
	Unchallenged	29°C	n/a	10	11	9.9	9.3	9.5	8.9	7.1	9.3	9.2	8.7	5.9	9.3	8.5	9.1	9.4	7.7
Viruses	Drosophila C virus	25°C	2000 TCID50	10	5	8.3	6.7	7.7	6.8	7.8	8.1	6.3	6.3	5.5	5.0	5.1	6.7	3.8	4.1
	Flock House virus	25°C	250000 TCID50	10	5	8.5	7.1	7.2	7.3	7.3	7.7	6.9	6.2	5.1	6.0	4.9	6.8	5.2	5.1
	Drosophila X virus	25°C	431000 TCID50	10	2	9.0	6.9	8.2	7.1	7.9	7.8	5.4	7.4	4.9	5.1	6.0	6.2	5.6	4.6
	Invertebrate iridescent-6 virus	25°C	31700 TCID50	10	3	9.4	7.5	9.4	8.1	8.4	8.8	5.6	6.0	5.5	5.8	6.1	8.3	5.9	4.4
	Sindbis virus	25°C	30000 TCID50	10	3	9.4	8.4	8.6	8.9	8.8	8.3	5.4	5.9	4.9	4.8	4.5	7.0	4.5	4.3
Fungi	Aspergillus fumigatus (natural)	25°C	infect on plate	10	5	10.0	9.2	9.9	8.8	7.2	7.6	9.1	8.2	6.8	8.6	7.7	8.7	6.9	4.7
	Beauveria bassiana (natural)	29°C	30mg	10	5	9.6	8.2	9.8	9.4	7.3	7.8	8.7	7.7	5.0	6.0	6.9	6.4	5.0	4.4
	Beauveria bassiana	29°C	OO-5	7	6	6.4	6.0	3.6	5.7	3.6	5.2	5.3	4.4	2.6	3.4	4.4	2.1	3.1	3.3
	Candida albicans	29°C	OO-200	7	8	6.7	5.6	5.0	5.9	5.0	5.4	5.9	6.4	3.7	3.1	5.7	3.6	5.9	3.2
Gram positive bacteria	Mycobacterium marinum	25°C	OO+1	7	6	6.1	6.0	6.5	6.5	5.9	5.2	5.8	5.7	4.1	5.8	5.1	5.1	6.3	3.9
	Corynebacterium diptheriae	25°C	OO+100	7	3	6.0	6.2	6.4	5.6	5.1	6.6	4.6	5.8	3.2	3.3	4.9	7.2	5.9	4.5
	Micrococcus luteus	29°C	OO-200	7	8	6.7	5.7	6.8	6.2	6.4	6.8	6.5	5.6	4.6	6.5	8.0	4.8	7.7	6.3
	Streptococcus pneumoniae	25°C	OO+0.3	7	3	6.8	6.1	6.4	6.7	6.1	6.6	6.3	6.4	5.7	5.0	5.1	6.0	6.0	5.5
	Enterococcus faecalis	25°C	OO+5	7	10	5.6	4.5	5.6	4.2	5.8	5.8	5.2	3.9	1.6	1.8	3.2	4.2	4.9	1.4
	Staphylococcus aureus	25°C	OO+0.5	7	8	5.1	4.8	5.7	4.2	5.8	3.4	3.6	1.9	2.0	2.1	2.1	4.1	3.9	3.4
	Listeria monocytogenes	25°C	OO+0.5	7	5	4.8	4.7	2.2	2.3	1.8	1.2	4.6	3.9	2.1	2.2	2.2	3.5	4.8	2.0
	Bacillus subtilis	25°C	OO+5	7	7	5.7	3.3	4.1	4.0	3.8	3.8	5.0	4.5	2.8	2.7	3.4	4.5	5.4	2.1
	Vibrio parahaemolyticus	25°C	OO+10	7	2	3.1	1.1	2.9	1.1	1.3	1.3	2.8	3.2	1.5	1.9	1.9	1.0	1.1	1.0
	Providencia nigres	25°C	OO+0.1	7	7	5.5	1.1	5.1	2.0	4.4	4.4	3.6	1.5	1.5	5.3	4.0	1.3	7.8	2.1
Gram negative bacteria	Providencia burfordgranaria	25°C	OO+10	7	7	5.1	1.7	4.6	1.5	4.6	4.8	4.9	4.2	2.9	4.9	4.0	1.3	7.8	1.0
	Pseudomonas aeruginosa	25°C	OO+10	7	8	6.4	1.5	5.8	1.8	5.8	6.8	6.4	5.2	4.7	5.2	4.0	1.8	7.2	1.4
	Klebsiella pneumoniae	25°C	OO+1	7	4	6.7	1.4	6.7	1.4	6.7	6.8	6.9	6.3	3.9	6.9	4.0	1.3	7.9	2.0
	Enterobacter cloacae	25°C	OO+10	7	8	6.8	1.3	6.7	1.0	6.8	6.6	6.7	6.1	4.9	6.1	5.8	1.3	7.9	1.0
	Escherichia coli	25°C	OO-200	7	8	6.6	5.0	7.9	3.8	6.2	7.0	5.4	5.5	3.9	5.3	4.0	3.8	4.0	1.7
	Salmonella enterica ser. typhimurium	25°C	OO-200	7	4	6.6	4.8	6.7	5.3	6.3	6.4	6.4	6.6	5.5	4.8	4.7	4.7	3.9	2.7

Figure 3

Heatmap of lifespans of immune module-deficient flies upon infection by various pathogens.

Darker blue indicates lower survival, while white indicates maximum survival. Experiments used either 7 or 10 days as a maximum lifespan/time course, and the heatmap is adjusted accordingly per row to have white fill for the maximum possible lifespan of that row. Heatmap colours: blue (low survival), white (high survival). Survival curves are presented in Supplementary file 2, and a summary of susceptibilities is provided in Figure 3supp1. Small numbers beside mean lifespans indicate total sample size per genotype per treatment.

Surprisingly, our study reveals a relatively consistent role of melanization against all the tested viruses, notably so for DXV, IIV-6, and SINV. ΔMel flies also display a minor increased susceptibility to *B. bassiana* natural infection and septic injury. As previously described (Dudzic et al., 2019), ΔMel flies are highly susceptible to the Gram-positive bacterium *S. aureus*, rivaling the susceptibility of $\Delta ITPM$ flies. ΔMel flies further show a strong susceptibility to *Pr. rettgeri* rivalling ΔIMD flies, and also a minor susceptibility to infection by other Gram-negative bacteria (*Pr. burhodogranariae*, *E. coli*).

Collectively, the use of single module mutants confirms the major role of the Imd pathway against Gram-negative bacteria and Toll against virulent Gram-positive bacteria and fungi. Interestingly, melanization was consistently important to survive virus infection, and plays a prominent and important role in defense against specific bacterial species. While it was almost never the most critical module, phagocytosis and the cellular response contributes to survival against various germs. Use of single module deficient flies alongside $\Delta ITPM$ flies reveals that survival to some of the microbes tested can rely almost entirely on one pathway (Imd – *Ecc15*, *K. pneumoniae*, *E. cloacae*, *Pr. burhodogranariae*; Toll – *E. faecalis*), or can receive notable contributions from two pathways (Toll and IMD – *L. monocytogenes*, *B. subtilis*, *V. parahemolyticus*; Toll and Phagocytosis – *A. fumigatus* and *B. bassiana* natural infection; Imd and Melanization – DCV). Strikingly, many pathways contribute to host defense to *S. aureus* and *Pr. rettgeri*.

Contribution of AMPs and Bomanins to Toll and Imd activities

Recent studies have described a prime importance of antimicrobial peptides (AMPs) and host defense peptides (HDPs) in responding to infection (reviewed in Hanson (2024)). These peptides are the principal effectors of the Toll and Imd pathways, yet mutations for AMPs and HDPs typically have not been tested alongside mutations for both the Imd and the Toll pathways across pathogens, as studies have typically included just the one relevant pathway mutation as a positive control. Thus, the importance of AMPs and HDPs relative to the pathways that regulate them has been only partially investigated. We used $\Delta AMP14$ and ΔBom flies to assess the extent by which AMPs and Bomanins contribute to Toll and Imd activities (Figure 3). Here we found that AMPs are critical for survival to infection against all Gram-negative bacteria tested, often rivaling the susceptibility of ΔIMD flies. Notably, $\Delta AMP14$ flies were also somewhat susceptible to the yeast *C. albicans*, and the Gram-positive bacteria *B. subtilis* and *M. luteus*, reinforcing minor susceptibilities seen previously (Carboni et al., 2022; Hanson et al., 2019). Interestingly, we found that $\Delta AMP14$ survivals to DCV, FHV, DXV, and IIV6, largely paralleled those of ΔIMD flies (Figure 3, Supplemental File 1), suggesting Imd-mediated survival phenotypes rely greatly on the presence of the seven classical AMP families.

On the other hand, ΔBom flies were susceptible to septic injury by *B. bassiana* and also somewhat to *C. albicans* consistent with previous studies (Supplementary file 2), as well as all virulent Gram-positive bacteria, consistent with their important role downstream of the Toll pathway (Clemmons et al., 2015; Lindsay et al., 2018; Xu et al., 2023). In addition, ΔBom flies paralleled the $\Delta TOLL$ susceptibility to DCV, FHV, and DXV, but were not susceptible to IIV6 like $\Delta TOLL$.

Collectively, our study confirms the key role of antimicrobial peptides downstream of the Imd pathway in the defense against Gram-negative bacteria, and a primary role of Bomanins downstream of Toll in defense against fungi and Gram-positive bacteria, with little input to survival against Gram-negative bacteria. We additionally find these secreted peptides can contribute to survival after certain viral infections.

Immune modules mostly contribute additively to host defense

The results described above revealed the key contribution of single modules and immune effectors to host defense. However, they did not assess the possible additive, synergistic, or antagonistic contribution of these four modules. Moreover, they did not monitor to which extent these modules

were important in the absence of other modules. To address this, we compared the survival of single-module deficient flies to double module-deficient flies lacking two modules, and $\Delta ITPM$ flies lacking all four modules (**Figure 3** [↗](#)).

We first observed that $\Delta ITPM$ flies, which lack the four modules, were always as or more susceptible than single and double module deficient flies. In multiple cases, single or double mutants already rivalled the susceptibility of $\Delta ITPM$ flies in microbe-specific ways. In particular, the susceptibility of ΔIMD -flies to most Gram-negative bacteria rivalled $\Delta ITPM$, and for some germs $\Delta TOLL$ (*L. monocytogenes*, *E. faecalis*, *B. bassiana* injury) or ΔMel (*S. aureus*) alone were comparable to $\Delta ITPM$. In other cases, however, combined loss of two pathways was necessary to rival $\Delta ITPM$ susceptibilities. For instance, ΔIMD , $\Delta TOLL$ deletion causes complete $\Delta ITPM$ -like susceptibility to *B. subtilis*, *M. luteus*, and *E. coli*, and near-complete susceptibility to *Sa. typhimurium*. ΔIMD , $\Delta TOLL$ further causes increased susceptibility to most viruses compared to their individual mutant strains. In these cases, there is clear synergy or additivity of the contributions of the two pathways, as either pathway alone results in only a minor or no susceptibility. We also observed intriguing differences between our two versions of $\Delta TOLL$, ΔMel flies, as $\Delta TOLL$, ΔMel^{PPS} tended to succumb to infection to a greater extent than $\Delta TOLL$, ΔMel^{HP} , consistent with the differences observed upon clean injury (**Figure 2** [↗](#)).

$\Delta TOLL$, ΔMel^{PPS} displayed a minor increase in susceptibility compared to $\Delta TOLL$, ΔMel^{HP} to *A. fumigatus* natural infection, and more prominent susceptibilities to the yeast *C. albicans*, the Gram-positive bacteria *M. marinum*, *M. luteus* and also to Gram-negative bacteria *K. pneumoniae*, *E. coli*, *Pr. rettgeri* and *Pr. burhododranariae*.

Collectively, use of double module deficient flies uncovers the role of immune modules to host defense that were masked by other modules. While ΔIMD and $\Delta TOLL$ deficiencies sometimes displayed a synergistic effect (i.e. little mortality of single mutants, but $\Delta ITPM$ -like mortality of double mutants), for combinations of other pathways the results were more often minor with a cumulative effect on susceptibility, suggesting most modules contribute to defence independently. The use of two variations of $\Delta TOLL$, ΔMel , with $\Delta TOLL$, ΔMel^{PPS} having a stronger impact than $\Delta TOLL$, ΔMel^{HP} , also highlights how the level of pathway disruption can affect infection outcomes, particularly for certain fungal or bacterial pathogens like *Providencia* species that may cleave host proteases (suggested by [Duneau et al., 2017b](#) [↗](#); [Issa et al., 2018](#) [↗](#)).

Timing of Toll, Imd, Phagocytosis, and Melanization contribution to host defense

Host defense programs can rely on resistance mechanisms that directly target or limit growth of pathogens ([Howick and Lazzaro, 2017](#) [↗](#); [Lafont et al., 2021](#) [↗](#); [Medzhitov et al., 2012](#) [↗](#)). When effectors reach a critical concentration threshold that promotes resistance, pathogen growth is inhibited; for instance, the time to pathogen control of *Pr. rettgeri* is ~7 hours, relying heavily on the expression and production of DptA ([Duneau et al., 2017a](#) [↗](#); [Hanson et al., 2023](#) [↗](#)). Phagocytosis and the melanization response are often described as providing near-immediate immune protection ([Haine et al., 2008](#) [↗](#)), with reactions ex vivo progressing within minutes. Meanwhile activation of the Imd and Toll pathways takes hours to reach peak concentrations of host defense peptides ([Uttenweiler-Joseph et al., 1998](#) [↗](#)). Thus, we were curious if different mutations would display differences in their time of action to control microbial growth. We therefore measured the microbial growth rate for *Pr. rettgeri*, *S. aureus*, and *C. albicans* at different time points post-infection in wild-type and single module mutant flies as these pathogens were combatted by several immune modules in our survival data.

Agreeing with contributions to survival, flies singly deficient for IMD, Phagocytosis, or Melanization modules display a higher bacterial load than wild-type when infected with *Pr. rettgeri* (**Figure 4A** [↗](#)), which was apparent after 12 hours post-infection. Our observation that

Δ Phag showed impaired IMD signalling at 6h (**Figure 1A**) might explain the contribution of phagocytosis to *Pr. rettgeri* microbial growth control with the same timing as Δ IMD flies. We observed that Δ IMD flies, Δ Mel flies, and Δ ITPM flies that each succumb completely to *Pr. rettgeri* infection, also show high and consistent microbe loads at 24h indicative of sepsis-induced death. On the other hand, wild-type, Δ TOLL, and Δ Phag flies show a greater stochasticity in bacterial load at 24h, consistent with a fraction of individuals surviving the infection by controlling the pathogen. This indicates deletion of the Toll pathway and phagocytosis does not fully ablate essential resistance mechanisms for combatting *Pr. rettgeri*.

In the case of *S. aureus* (**Figure 4B**), we observed some individuals with higher bacterial loads in Δ Phag, and Δ IMD flies at 2h and 6h, which was not seen in wild-type, Δ TOLL, or Δ Mel flies. This suggests the Imd pathway can suppress early *S. aureus* growth, and possibly also phagocytosis, although this Δ Phag could ultimately stem from lesser Imd induction. Interestingly, flies lacking melanization displayed the highest susceptibility, but did not depart from wild-type or other modules in pathogen load until 12h post-infection. While the melanization reaction is a rapid response, these results suggest that the killing activity of the melanization response acts with slower kinetics. In Δ ITPM flies, we further observed earlier *S. aureus* growth at 2h and 6h like Δ IMD and Δ Phag flies. However, *S. aureus* loads at 12-24h were more comparable in Δ ITPM flies to Δ Mel flies. These trends agree both with survival data kinetics, and an independent action of each of these pathways in their resistance effects against *S. aureus*. This suggests that Δ ITPM flies succumb even more quickly than individual modules due to loss of multiple resistance mechanisms with different kinetics of activation that independently contribute to defense.

For *C. albicans* (**Figure 4C**), mortality begins 2-3 days post-infection for Δ TOLL mutants, but takes place at later time points for Δ IMD, Δ Phag, and Δ Mel flies. In microbial growth kinetics, Δ TOLL already shows high *C. albicans* loads at 24h consistent with a previous study (Hanson et al., 2019) and observations done with *Candida glabrata* (Quintin et al., 2013). Interestingly, a few Δ IMD flies also showed elevated growth of *C. albicans* at early time points. This suggests Imd-regulated effectors could help to suppress initial *C. albicans* growth, while Toll-regulated genes (e.g. Bomanins, Drs) contribute to this pathogen suppression to a greater extent at later time points. On the other hand, Δ Phag, and Δ Mel flies did not show any increased *C. albicans* loads within the 24h time window despite onset of mortality at later time points. Finally, Δ ITPM flies displayed more consistent and rapid growth of *C. albicans* than Δ TOLL or Δ IMD alone, particularly visible at 12-24h post-infection, consistent with independent contributions of both Toll and Imd in resistance to this yeast.

All survival experiments to this point were done with males. We therefore assessed key survival trends for these infections in females to learn whether the dynamics we observed were consistent across sexes (**Figure 4supp1**). For all three pathogens (*Pr. rettgeri*, *S. aureus*, *C. albicans*) the rank order of susceptibility was broadly similar between males and females, with higher rates of mortality in females overall. Thus, we found no marked sex-by-genotype interaction. Interestingly, the greater susceptibility of females in our hands is true even for Δ ITPM flies, although there were only a few surviving flies on which we can base these conclusions. However, these data may suggest the sexual dimorphism in defense against infection that we see against these pathogens is due to factors independent of the immune modules we disrupted.

Collectively, we observed that microbial growth parallels the susceptibility of modules indicating that they work in resistance. Surprisingly we did not observe an early role in pathogen killing for melanization despite melanization regulating a rapid blackening response in ex vivo assays (Nakhleh et al., 2017). We further recover stochasticity in microbe loads across genotypes as expected when only a subset of individuals suppress the pathogen and survive the infection (Duneau et al., 2017a). The more rapid and more consistent microbial growth kinetics of Δ ITPM flies, particularly visible by 18h, further demonstrate that these pathways contribute to resistance collectively.

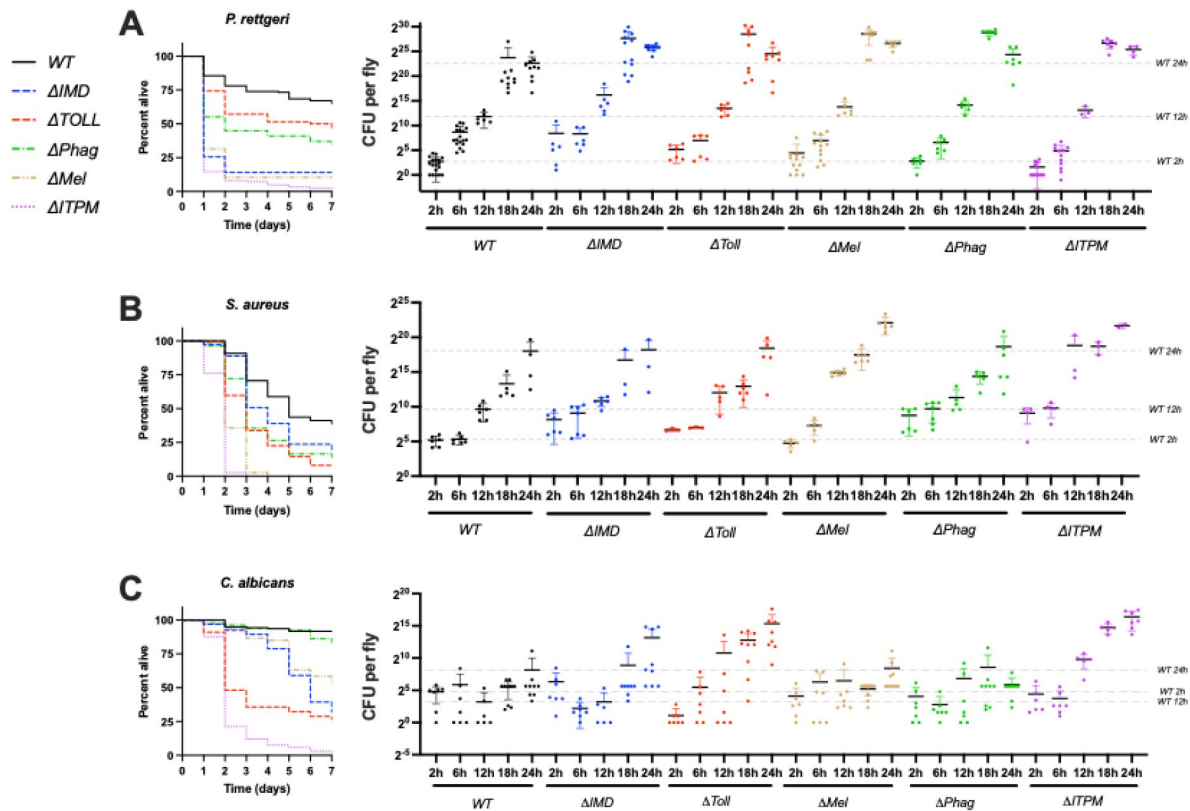


Figure 4

Growth kinetics of *Pr. rettgeri* (A), *S. aureus* (B), and *C. albicans* (C) in wild-type, single module mutant and $\Delta ITPM$ flies.

Survival curves underlying data in [Figure 3](#) are shown for comparison. Each data point reflects a pooled sample of 5 flies. Error bars reflect 1 standard deviation from the mean.

Contributions of individual modules to disease tolerance

An increasingly used metric to delineate roles of resistance and tolerance is the Pathogen Load Upon Death (PLUD) (Duneau et al., 2017a [↗](#); Duneau et al., 2025 [↗](#)). The PLUD is determined by the virulence of the pathogen to set an upper limit, but also the tolerance of the host in surviving to a given pathogen burden before succumbing to infection. We monitored PLUDs for the same pathogens: *Pr. rettgeri*, *S. aureus*, and *C. albicans*. The PLUD of both *Pr. rettgeri* and *S. aureus* upon death was overall similar in wild-type and single-module mutant flies (**Figure 5A-B** [↗](#)). Notably, there was a greater stochasticity in *Pr. rettgeri* infections trending towards lower PLUD in $\Delta ITPM$ flies, suggesting a stochastic reduction in disease tolerance of $\Delta ITPM$ flies to this bacterium, although this was not significant compared to wild-type ($P > .05$). We saw no statistical differences in PLUD across genotypes for *S. aureus*.

For *C. albicans*, the distribution of PLUD values for $\Delta Phag$ flies was markedly different from wild-type, also seen to some extent for ΔMel . Interestingly, flies with these genotypes suppress *C. albicans* growth well (**Figure 4C** [↗](#)). Yet here we recovered many individuals from both of these module-deficient lines that died with a far lower PLUD (**Figure 5C** [↗](#)). Taken together, this suggests that the humoral immune pathways, particularly Toll, contribute to resistance against *C. albicans*. However, the cellular response and melanization reaction instead regulate tolerance to *C. albicans* infection. The use of $\Delta ITPM$ flies further emphasizes that when both resistance (Toll, Imd) and tolerance (Phagocytosis, Melanization) mechanisms are deficient, the loss of resistance plays a primary importance, and can mask the loss of tolerance from being observed.

Collectively, monitoring of PLUD suggests that some modules, specifically phagocytosis and melanization, contribute to tolerance of certain infections. Importantly, this tolerance effect was only seen in genotypes that can resist infection. These findings demonstrate the collective contribution to resistance and/or tolerance of the four immune modules studied here.

Discussion

In this article, we have generated a set of single and compound immune module deficient fly lines in a controlled genetic background. Using various assays, we confirmed the validity of our lines, revealing that each module can be activated independently. We did, however, observe a higher Toll activation in ΔMel flies upon systemic infection with dead bacteria. Future studies may reveal if this higher Toll activity in ΔMel flies involves a regulatory pathway, or perhaps reflects higher persistence of microbial elicitors (e.g. peptidoglycan) that activate the Toll pathway. We also observed a lesser Imd activation in $\Delta Phag$ flies at 6 hours post-infection, which could reflect a role for hemocytes to stimulate the Imd systemic response. In this study, we used *NimC1¹; eater¹* double mutants to assess the cellular response. These flies have defects in phagocytosis, but also adhesion and sessility (Melcarne et al., 2019 [↗](#)). *NimC1¹; eater¹* larvae also have increased hemocyte number at the larval stage, but adults rapidly lose Hml positive cells (Melcarne, 2020 [↗](#)). Despite these limitations, we believe that the mutations we have used here are among the best available to assess the role of these four modules to host defense. Surprisingly, we could produce a fly line lacking the four main defense mechanisms of the systemic immune response, indicating that none of these modules is essential for survival. Theoretically, $\Delta ITPM$ flies are almost completely immune deficient but they can still clot wounds, activate the JNK and JAK-STAT pathways that mediate the wound healing response, and retain constitutive immune defense molecules that could also provide a certain degree of protection. Consistent with several studies (Capilla et al., 2017 [↗](#); Carvalho et al., 2014 [↗](#); R  met et al., 2002 [↗](#)), we show that Toll and melanization synergistically contribute to wound healing in adults. However, the observation that $\Delta Phag$, ΔMel or ΔIMD , ΔMel double mutant flies are also more susceptible to clean injury than ΔMel flies indicates that the Imd pathway and phagocytosis also contribute to wound healing. Thus,

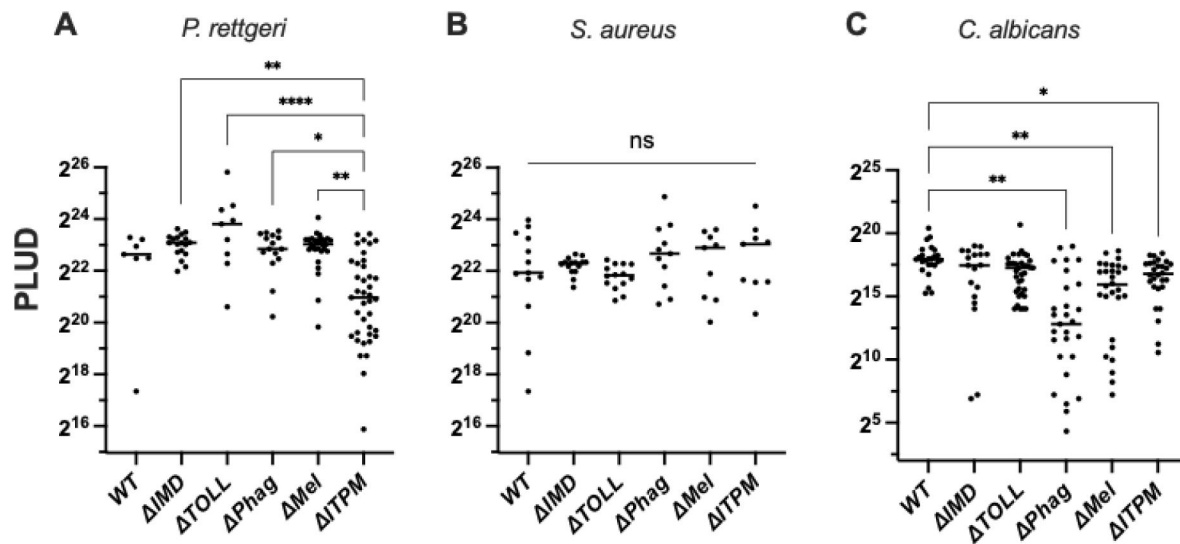


Figure 5

Measurement of PLUD upon infection with *Pr. rettgeri*, *S. aureus*, and *C. albicans*.

A) The PLUD of *Pr. rettgeri* in individual module mutant flies is not significantly different from wild-type. The PLUD of *ΔITPM* flies was not significantly different, although censoring of a single low-PLUD outlier in the wild-type would result in a significant difference between wild-type and *ΔITPM* flies ($p < .01$). **B)** The PLUD of *S. aureus*-infected flies is not different across any genotype. **C)** The PLUD of *C. albicans*-infected flies was significantly lower in *ΔPhag* flies with a notably lower mean PLUD. This difference was robust to use of different *ΔPhag* genetic backgrounds (merged data shown here). *ΔMel* flies and *ΔITPM* flies also had significantly lower PLUD. Error bars reflect 1 standard deviation from the mean.

although immune deficient fly lines for the four modules are viable, the use of compound mutation and *ΔITPM* flies reveals a clear role of the immune system in response to wounding and lifespan maintenance.

Use of single and double module-deficient flies provides key insights on the mechanisms used by *Drosophila* to combat infection. We could confirm previous studies revealing the role of Imd against Gram-negative bacteria, Toll against Gram positive bacteria and virulent fungi, and an importance of melanization and phagocytosis against specific pathogens (Charroux and Royet, 2009 [↗](#); Defaye et al., 2009 [↗](#); Garg and Wu, 2014 [↗](#); Nehme et al., 2011 [↗](#)). Surprisingly, melanization was a consistently important module in survival to virus infection. It seems unlikely that prophenoloxidase activity taking place in the hemolymph could combat viral agents that are intracellular. We instead speculate this contribution of melanization to surviving virus infection could be due to its role in wound healing, or perhaps autotoxic contributions of melanization reaction intermediates that fail to convert in phenoloxidase-deficient flies. We additionally recovered a role of Imd signalling (i.e. *Relish*) in antiviral defense, which was expected given recent characterizations of cGRLR-Sting-Relish antiviral immunity (Ai et al., 2024 [↗](#); Cai et al., 2022 [↗](#); Goto et al., 2018 [↗](#)). However, susceptibilities of *Rel^{E20}* flies were paralleled by *ΔAMP14* in all cases, suggesting AMPs contribute to this susceptibility. While Sting regulates a number of genes likely important for antiviral defense (Goto et al., 2018 [↗](#)), the susceptibility we observe here could be a direct action of AMPs on enveloped viruses as described in some studies (Feng et al., 2020 [↗](#); Huang et al., 2013 [↗](#); Yasin et al., 2004 [↗](#)), or an indirect effect, such as a failure to regulate gut microbes after the damage induced by viral replication (Marra et al., 2021 [↗](#)). We additionally used dual modes of infection for the fungus *B. bassiana*, finding an importance of melanization and Toll signalling in both infection modes. However, our study reveals that Bomanin effectors explain most of the Toll contribution upon septic injury but not natural infection, for *B. bassiana*. This observation is in line with different importance of effectors or modules according to the route of infection (Martins et al., 2013 [↗](#)).

Our double module mutant analysis reveals that most modules contribute additively to host defense. This is consistent with these modules functioning independently. However, we noticed instances of synergy between two pathways, notably including Toll and Imd. Synergy between Toll and Imd can be explained by the fact that many immune-inducible regulated genes receive input from both the Imd and Toll pathways, including genes like *Metchnikowin*, *Drosomycin* and *Transferrin1* (De Gregorio et al., 2002 [↗](#)). Finally, we also observed rarer cases of synergistic susceptibility in flies deficient for Toll and Melanization or Imd and Melanization.

Our study confirms that the Toll and Imd humoral modules provide a broad role against certain classes of pathogens, Imd for Gram-negative bacteria and DAP-type containing Gram positive bacteria, and Toll for Fungi and Gram-positive bacteria. Use of AMP and Bomanin mutants revealed that this can be largely explained by the effectors they control. In contrast, the contribution of phagocytosis and melanization appears to be critical to a more specific and diverse set of pathogens. We speculate that phagocytes or melanization are critical to handle bacteria that resist host defense peptides (Hanson et al., 2019 [↗](#)), or can hide from them (Touré et al., 2023 [↗](#)). The melanization reaction is a source of ROS that is potent against pathogens such as *S. aureus* that have been shown to be sensitive to ROS, and resistant to Toll and Imd defenses (Dudzic et al., 2019 [↗](#); Ford and King, 2021 [↗](#); Needham et al., 2004 [↗](#); Ramond et al., 2021 [↗](#)). These two modules play important roles in immune related processes such as encapsulation (melanization), the uptake of bacteria escaping from the gut, or tissue homeostasis (phagocytosis) (Braun et al., 1998 [↗](#); C. Melcarne et al., 2019 [↗](#); Nehme et al., 2007 [↗](#)) that were not assessed here. Collectively, our study validates, with minor discrepancies, many studies that have assessed the contribution of these modules individually (ex. (Apidianakis et al., 2005 [↗](#); Binggeli et al., 2014 [↗](#); Lamiable et al., 2016 [↗](#); Lemaitre et al., 1996 [↗](#), 1995 [↗](#); Nehme et al., 2011 [↗](#)). Our mutant lines can now be used

to analyze the contribution of these immune modules in resistance to other pathogens, notably wasps, nematodes, microsporidia and protozoans or in other contexts such as mating and local infection.

While some immune modules play a predominant role against some pathogens, other pathogens are handled by multiple modules. We hypothesize that pathogens that have intermediate levels of virulence, killing only a fraction of wild-type flies, may better reveal the role of multiple modules. Indeed, the stochasticity in survival analysis partly stems from the arms-race occurring between the pathogen and host immunity, as shown for *Pr. rettgeri* (Duneau et al., 2017a [↗](#)). In this condition, any small change in the immune system may tip the outcome of the arm-race toward lethality or survival. In this line, it is notable that multiple modules were important to survive *Pr. rettgeri* infection. Previous studies have revealed a major role of the Imd pathway-regulated AMP Diptericin A against this bacterium (Hanson et al., 2023 [↗](#), 2019 [↗](#); Unckless et al., 2016 [↗](#)). However, Duneau et al. showed that survival patterns to *Pr. rettgeri* bifurcate into two outcomes based on time taken to fully activate systemic defenses (Duneau et al., 2017a [↗](#)), and showed a role for a Toll-PO SP cascade regulating serine protease in defense against this microbe (Duneau et al., 2017b [↗](#)). We may speculate that melanization, although not as uniquely critical as Diptericin, might tip the balance of this arms race towards host lethality, resulting in comparable survival phenotypes. The observation that Δ Phag flies expressed less *DptA* (Figure 1 [↗](#)) can also explain their susceptibility. Future studies will clarify how these multiple modules can contribute to host survival according to additive or achilles dynamics – the concept that microbes have generic or specific weaknesses that host effectors target (Hanson, 2024 [↗](#)). It will also be important to consider the distinct roles of resistance, tolerance, and resilience in host defense (Howick and Lazzaro, 2017 [↗](#); Wukitch et al., 2023 [↗](#); Duneau et al., 2025 [↗](#)).

We observed a good correlation between survival analysis and pathogen growth in single module flies for *Pr. rettgeri*, *S. aureus*, and *C. albicans*. This indicates that these immune modules mostly contribute to resistance mechanisms that target pathogen growth. Our study did not reveal key early contributions of phagocytosis or melanization to control pathogen growth, despite their quasi-immediate activation; higher *S. aureus* growth at 2 hours in Δ Phag flies was paralleled by Δ Imd flies, and we observed lesser Imd activation in Δ Phag flies. Melanization, while being critical to survive *S. aureus* infection, impacts bacterial growth beginning only at the 12 hour time point. This indicates that the Melanization microbicidal activity in vivo takes place slower than the blackening reaction seen from bled hemolymph. Interestingly, Δ Phag and also Δ Mel flies could suppress *C. albicans* yeast growth, but ultimately some individuals succumb to infection with lower PLUD levels. We confirmed Δ Phag PLUD results using both an isogenic and a second wild-type genetic background, suggesting this lower PLUD is genuine. Susceptibility to fungal infection independent of fungal proliferation has also been reported using an *A. fumigatus* septic infection model, and relies on the protection offered by Bomanins from pathogen-secreted toxins (Xu et al., 2023 [↗](#)). It is tempting to speculate, based on the modules involved, that the loss of tolerance to *C. albicans* we observe has to do with wound repair or accumulating damage, such as what has been reported in beetles that suffer renal failure (Khan et al., 2017 [↗](#); Li et al., 2020 [↗](#)), or flies with autotoxic trachea degradation upon stress pathway disruption (Rommelaere et al., 2024 [↗](#)). Thus, this tolerance effect could rely on pathogen-mediated or autotoxic damage, which may be elucidated in a future study.

Here we have provided a single and double mutant analysis of *Drosophila* immune module functions. Our study provides several insights on what modules are most important to survive infection by defined pathogens. However, we also highlight the collective contribution of modules to defence even when one module is of an outsized importance. We extend our comprehension of innate immune responses by revealing higher complexity, implicating multiple host defense modules in survival to various germs, including some with more cryptic contributions. As illustrated by our previous characterizations of AMP function (Carboni et al., 2022 [↗](#); Hanson et al., 2019 [↗](#)), the melanization response (Dudzic et al., 2015 [↗](#)), and stress-induced Turandot

proteins (Rommelaere et al., 2024), a combinatorial mutation approach to deciphering immune functions can be extended even to the broad level of immune modules. Of note, we were unable to systematically sample all genotype-by-pathogen interactions equally. We have therefore been highly conservative in our reporting of major effects. There are likely many important interactions not discussed in our study. Future investigations may highlight important biology that is apparent in our data, but which we may not have mentioned here. To this end, we have deposited our isogenic immunity fly stocks in the Vienna Drosophila Resource Centre to facilitate their use. Beyond immunity, our tools can also be of use to study various questions at the cutting edge of aging, memory, neurodegeneration, cancer, and more, where immune genes have been implicated repeatedly. We hope that this set of lines will be useful to the community to better characterize the *Drosophila* host defense.

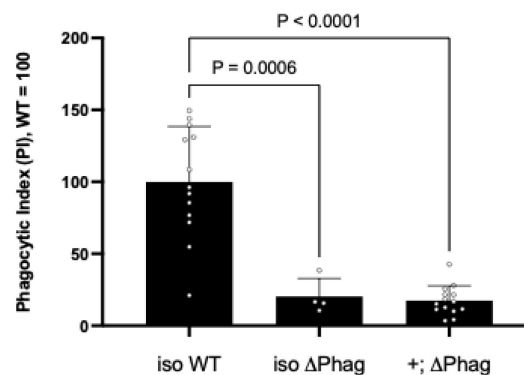


Figure 1supp1

We used $\Delta Phag$ mutants in the genetic background of Melcarne et al. (2019) (+; $\Delta Phag$) in some experiments, and confirm here that these mutants are equally deficient in phagocytic capacity to $\Delta Phag$ flies from the iso DrosDel genetic background ($\Delta Phag$).

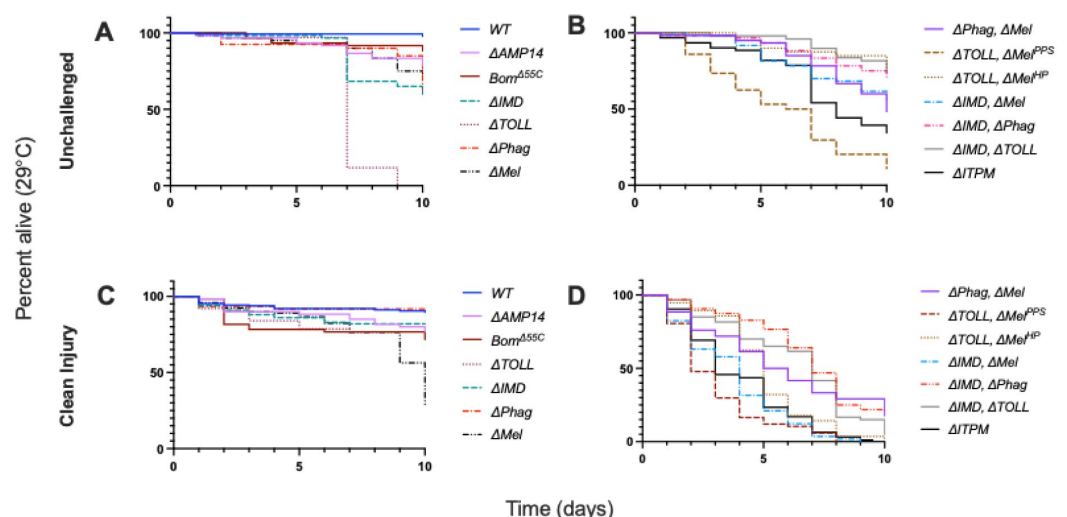


Figure 2supp1

Lifespans of mutants used in this study in unchallenged flies and upon clean injury conditions at 29°C.

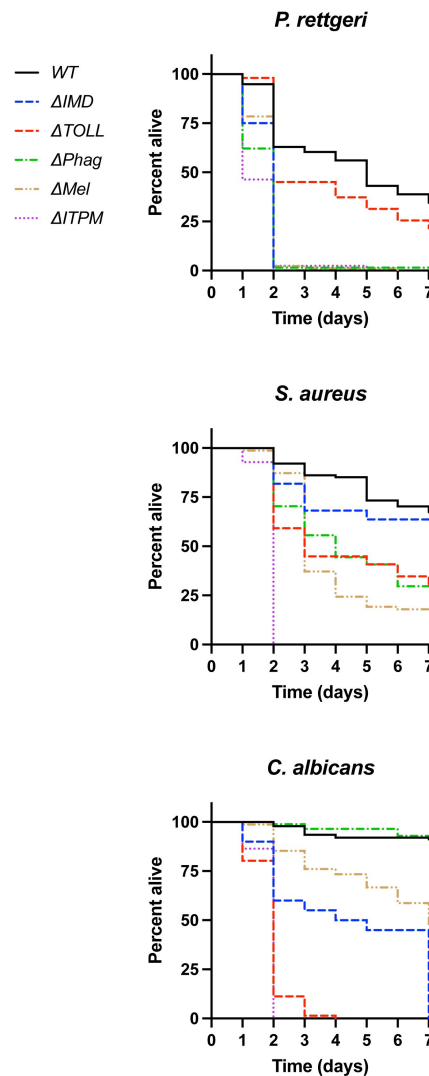


Figure 4supp1

Survival to infection of females against *P. rettgeri* (A), *S. aureus* (B), and *C. albicans* (C).

Survival trends largely parallel the rank order of males (Figure 4 [↗](#)).

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Additional information

Author contributions

Conceptualization: FR, MAH, BL Data curation: FR, MAH

Funding acquisition: BL, MAH Investigation: FR, YT, MR, FS Methodology: FR

Project administration: BL Validation: YT, MAH Resources: FR, MAH Supervision: MAH, BL

Writing – original draft: FR, BL Writing – review & editing: MAH, BL

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Additional files

Supplementary file 1. *Supplemental discussion of fly genetics.* 

Supplementary file 2. *Survival curves for all data summarized in Figure 3.* 

Supplementary file 3. *Supplemental discussion of fly genetics and considerations for the use of genotypes described in this study.* 

Figure 3supp1. *Summary table of susceptibilities per the three conditions outlined in section “Systemic infections and survival.” A ✓ indicates the pathway contributes to defence against a given pathogen. In the event that a single module mutant is not more susceptible to the infection, but contributes to increased mortality in a double mutant line, a ✓ is given and the co-deleted relevant pathways highlighted by their first letter. Black boxes indicate no difference to wild-type or to clean injury.* 

Table S1. *Microbe characteristics and growth conditions.* 

Note

This reviewed preprint has been updated to improve the resolution of Table 1 and Figure 3.

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Reviewer #1 (Public review):

Summary:

The innate immune system serves as the first line of defense against invading pathogens. Four major immune-specific modules-the Toll pathway, the Imd pathway, melanization, and phagocytosis-play critical roles in orchestrating the immune response. Traditionally, most studies have focused on the function of individual modules in isolation. However, in recent years, it has become increasingly evident that effective immune defense requires intricate interactions among these pathways.

Despite this growing recognition, the precise roles, timing, and interconnections of these immune modules remain poorly understood. Moreover, addressing these questions represents a major scientific undertaking.

Strengths:

In this manuscript, Ryckebusch et al. systematically evaluate both the individual and combined contributions of these four immune modules to host defense against a range of pathogens. Their findings significantly enhance our understanding of the layered architecture of innate immunity.

<https://doi.org/10.7554/eLife.107030.2.sa2>

Reviewer #2 (Public review):

Summary:

In this work, the authors take a holistic view at the *Drosophila* immunity by selecting four major components of fly immunity often studied separately (Toll signaling, Imd signaling, phagocytosis and melanization), and studying their combinatory effects on the efficiency of the immune response. They achieve this by using fly lines mutant for one of these components, or modules, as well as for a combination of them, and testing the survival of these flies upon infection with a plethora of pathogens (bacterial, viral and fungal).

Strengths:

It is clear that this manuscript has required a large amount of hands-on work, considering the number of pathogens, mutations and timepoints tested. In my opinion, this work is a very welcome addition to the literature on fly immune responses, which obviously do not occur one type of a response at a time, but in parallel, subsequently and/or are interconnected. I find that the major strength of this work is the overall concept, which is made possible by the mutations designed to target the specific immune function of each module, without effects on other functions. I believe that the combinatory mutants will be of use for the fly community and enable further studies of interplay of these components of immune response in various settings.

To control for the effects arising from the genetic variation other than the intended mutations, the mutants have been backcrossed into a widely used, isogenized *Drosophila* strain called w1118. Therefore, the differences accounted for by the genotype are controlled.

I also appreciate that the authors have investigated the two possible ways of dealing with an infection: tolerance and resistance, and how the modules play into those.

Weaknesses:

While controlling for the background effects is vital, the w1118 background is problematic (an issue not limited to this manuscript) because of the wide effects of the white mutation on several phenotypes (also other than eye color/eyesight). It is a possibility that the mutation influences the functionality of the immune response components. I acknowledge that it is not reasonable to ask for data in different backgrounds better representing a "wild type" fly, but I think this matter should be brought up and discussed.

The whole study has been conducted on male flies. Immune responses show quite extensive sex-specific variation across a variety of species studied, also in the fly. But the reasons for this variation are not fully understood. Therefore, I suggest that the authors would conduct a

subset of experiments on female flies to see if the findings apply to both sexes, especially the infection-specificity of the module combinations.

Comments on the revised manuscript:

I appreciate the author's responses to the points I raised and the additional work they have conducted. The authors have now discussed the possible background effect and added an experiment on female flies showing that the module function is applicable to both sexes.

<https://doi.org/10.7554/eLife.107030.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The innate immune system serves as the first line of defense against invading pathogens. Four major immune-specific modules - the Toll pathway, the Imd pathway, melanization, and phagocytosis- play critical roles in orchestrating the immune response. Traditionally, most studies have focused on the function of individual modules in isolation. However, in recent years, it has become increasingly evident that effective immune defense requires intricate interactions among these pathways.

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Strengths:

In this manuscript, Ryckebusch et al. systematically evaluate both the individual and combined contributions of these four immune modules to host defense against a range of pathogens. Their findings significantly enhance our understanding of the layered architecture of innate immunity.

We thank the reviewer for their kind assessment.

Weaknesses:

While I have no critical concerns regarding the study, I do have several suggestions to offer that may help further strengthen the manuscript. These include:

(1) Have the authors validated the efficiency of the mutants used in this study? It would be helpful to include supporting data or references confirming that the mutations effectively disrupted the intended immune pathways.

We have done so in Figure 1.

(2) Given the extensive use of double, triple, and quadruple mutants, a more detailed description of the mutant construction process is warranted.

We now provide a supplement (File S1) that details the successive genetic crosses and recombinations that were required to generate these compound fly stocks carrying multiple mutations. We also provide some information regarding rapid screening of stocks for

phenotypes. Of note some of these fly stocks have been deposited at VDRC as they will be useful to fly community to assess immune modules in a controlled background, and complete stock information will be tied to these stocks there.

Reviewer #2 (Public review):

Summary:

In this work, the authors take a holistic view of Drosophila immunity by selecting four major components of fly immunity often studied separately (Toll signaling, Imd signaling, phagocytosis, and melanization), and studying their combinatory effects on the efficiency of the immune response. They achieve this by using fly lines mutant for one of these components, or modules, as well as for a combination of them, and testing the survival of these flies upon infection with a plethora of pathogens (bacterial, viral, and fungal).

Strengths:

It is clear that this manuscript has required a large amount of hands-on work, considering the number of pathogens, mutations, and timepoints tested. In my opinion, this work is a very welcome addition to the literature on fly immune responses, which obviously do not occur in one type of response at a time, but in parallel, subsequently, and/or are interconnected. I find that the major strength of this work is the overall concept, which is made possible by the mutations designed to target the specific immune function of each module (at least seemingly) without major effects on other functions. I believe that the combinatory mutants will be of use for the fly community and enable further studies of the interplay of these components of immune response in various settings.

To control for the effects arising from the genetic variation other than the intended mutations, the mutants have been backcrossed into a widely used, isogenized Drosophila strain called w1118. Therefore, the differences accounted for by the genotype are controlled.

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While controlling for the background effects is vital, the w1118 background is problematic (an issue not limited to this manuscript) because of the wide effects of the white mutation on several phenotypes (also other than eye color/eyesight). It is a possibility that the mutation influences the functionality of the immune response components, for example, via effects of the faulty tryptophan handling on the metabolism of the animal.

I acknowledge that it is not reasonable to ask for data in different backgrounds better representing a "wild type" fly (however, that is defined is another question), but I think this matter should be brought up and discussed.

We agree with the reviewer and have included caveats on the different genetic effects brought about the combinatory mutant approach including differences in white gene status, insertion of GFP or DsRed markers, and nature of genetic mutations (Line 142-on).

“Of note, the strains used in this study differ in their presence/absence of the white⁺ gene, present in the PPO1^Δ, NimC1¹ and eater¹ mutations. In addition to its well established function in eye pigmentation, the white gene can also impact host neurology and intestinal stem cell proliferation (Ferreiro et al., 2017; Sasaki et al., 2021). We did not observe any obvious correlations between white⁺ gene status and susceptibilities in this study. Moreover, in a previous study looking at the cumulative effects of AMP mutations on lifespan, white gene status and fluorescent markers did not readily explain differences in longevity (Hanson and Lemaitre, 2023). We therefore believe that the extreme immune susceptibility we have created through deficiencies for pathways regulating hundreds of genes, or major immune modules, overwhelms the potential effects of white⁺ and other transgenic markers. For additional information on which stocks bear which markers, see discussion in Supplementary file 1.”

Of interest, we were highly conscious of this concern in working with combinatory AMP mutants which differed in white, GFP, and DsRed copies. However, even over the many weeks of snowballing effects on microbiota community composition and structure, we found no trends tied strictly to white⁺ or to other genetic insertions on lifespan (Hanson and Lemaitre, 2023; DMM).

The whole study has been conducted on male flies. Immune responses show quite extensive sex-specific variation across a variety of species studied, also in the fly. But the reasons for this variation are not fully understood. Therefore, I suggest that the authors conduct a subset of experiments on female flies to see if the findings apply to both sexes, especially the infection-specificity of the module combinations.

We thank the reviewer for this suggestion. We have performed the requested experiments, and include female survival trends in Figure 4supp1. We have added the following text to the main manuscript (Line 554):

“All survival experiments to this point were done with males. We therefore assessed key survival trends for these infections in females to learn whether the dynamics we observed were consistent across sexes (Figure 4supp1). For all three pathogens (Pr rettgeri, Sa aureus, C. albicans) the rank order of susceptibility was broadly similar between males and females, with higher rates of mortality in females overall. Thus, we found no marked sex-bygenotype interaction. Interestingly, the greater susceptibility of females in our hands is true even for ΔITPM flies, although there are only a few surviving flies on which we can base these conclusions. However, these data may suggest the sexual dimorphism in defense against infection that we see against these pathogens is due to factors independent of the immune modules we disrupted.”

It is worth noting that male-female sex dichotomies in infection are inconsistent across the literature, with strong lab-specific effects (Belmonte et al., 2020 and personal observation). In our lab setting, we consistently see female mortality higher than males when compared, independent of pathogen and mutant background. We have not seen notable interaction terms of sex and genotype for most immune deficient mutants. It is quite interesting to have done these experiments with ITPM, however, which reveals that there is at least a trend suggesting this dichotomy is independent of the four immune modules we deleted. Still, our infection conditions kill most males, and so it would be good to replicate this sex-specific ΔITPM result in a dedicated study with doses chosen to improve the resolution of male-female differences. For now, we prefer to use conservative language and avoid overinterpreting this trend, but do feel it merits mentioning.

Recommendations for the authors:

Comment on statistical requests

Both reviewers requested further clarity on the statistical analyses supplemental to Figure 3. We have addressed these comments as follows.

First, we now provide an additional supplementary .zip file containing summary statistics for all survival data in Figure 3 (Supplementary File 3). We have additionally added this text to line 226 to make this data treatment more clear:

...” we chose to focus on major differences apparent in summary statistics, highlighting”...

And we highlight that all survival data are also provided as Kaplan-Meier survival curves in the main or supplementary figures in Line 233:

“Kaplan-Meier survival curves for all experiments are provided in the main text or supplementary information”.

Second, as outlined in the main text, we were unable to sample across all pathogen-by-genotype interactions systematically, and this unfortunately obscures robust statistical modelling. We addressed the challenge of finding meaningful statistical differences by focusing on trends only if they were i) consistent across experimental replicates, ii) of a consistent logic across comparable genotypes, ensuring random inter-experimental noise was not unduly shaping interpretations, and iii) of a mean lifespan difference ≥ 1.0 days compared to wild-type, and compared to relevant unchallenged or clean-injury controls. This last choice was especially important because not all experimental replicates included all genotypes due to challenges of animal husbandry and coordination among multiple researchers over five years of data collection. As a result, our initial analyses using a cox mixed-effects model found it to be rather useless, being insensitive to important experiment batch effects visible to the eye because statistically-affected genotypes were not present in all experiments.

We therefore ensured that behaviour relative to controls within* experiments was consistent, rather than the comparison of genotypes to controls across the sum of experiments with a post-hoc treatment attempting to apportion variance to experiment batch (but unable to do so for some genotypes and some batches). Due to differences in baseline health and the dynamics explained by studies like Duneau et al. (2017; eLife, there is an expected unequal variance of genotype*pathogen interactions across experiment batches. Unfortunately, this unequal variance, coupled with incomplete sampling across experiment batches, means “highly significant” differences can emerge that don’t hold up to scrutiny of comparisons to controls taken only from within an experiment batch. Thus, we chose to forego a cox mixed effect model approach entirely. Instead, our highly conservative approach, focusing on only very large effects with a mean lifespan difference ≥ 1.0 days, mitigates these issues. We have taken great care to ensure that any results we highlight stand up to inter-experiment batch effects. We would further draw the reviewers’ attention to our response to Reviewer 2 relating to Figure 3, which emphasizes the level of conservatism that we are applying.

At the end of the Discussion, we have added the following sentence to emphasize these limitations:

“...a combinatorial mutation approach to deciphering immune function can be extended even to the broad level of whole immune modules. Of note, we were unable to systematically sample all genotype-by-pathogen interactions equally. We have therefore been highly conservative in our reporting of major effects. There are likely many important interactions” not discussed in our study. Future investigations may highlight important biology that is apparent in our data, but which we may not have mentioned here. To this end, we have

deposited our isogenic immunity fly stocks in the Vienna Drosophila Resource Centre to facilitate their use. Beyond immunity, our tools can also be of use to study various questions at the cutting edge of aging, memory, neurodegeneration, cancer, and more, where immune genes are repeatedly implicated. We hope that this set of lines will be useful to the community to better characterize the Drosophila host defense.”

We recognise this response may not fully satisfy the reviewers’ requests. While use of summary statistics is simple, our rules for highlighting interactions of importance are defined, readily understood and interpreted, and draw attention to key trends in that are backed by a solid understanding of the data and its limitations. We have taken this approach out of a responsibility to avoid making spurious assertions that stem from underpowered statistical models rather than from the biology itself.

Reviewer #1 (Recommendations for the authors):

(1) Lines 1092-1093 - Please double-check the labeling of the panels in Figure 2. It appears that panels A and C correspond to single-module mutants, whereas panels B and D refer to compound-module mutants.

We have modified Figure 2 and Figure 2supp1 labelling. We also realise there was an error in the column titling that contributed to the confusion. We hope the new layout is clear, and thank the reviewers for noting this issue.

(2) Lines 347-377 - Figure 2D is not cited in the text.

We now cite Fig2D in Line 356.

(3) P values should be indicated in Figure 2 and Figure 3 for all relevant comparisons. Additionally, "ns" (not significant) should be added in Figure 5A-B.

We make the effort to show key uninfected survival trends in Figure 2, and list the total flies (n_flies) in Fig3 to provide the reader with the underlying confidence in the trends observed. We focus on differences of mean lifespan of at least 1 day, and which are consistent in direction across combinatory mutations. We have avoided the multiple comparisons of cox proportional hazard survival analyses throughout this study because they are overly sensitive for our purposes, as we have previously when systematically comparing many genotypes to each other (see Hanson and Lemaitre, 2023; DMM).

(4) Minor points: Hml-Gal4, UAS-GFP should be italic; Line 192-- "uL" and "uM"; Line 596: P>.05.

We have made these changes. We’re unsure what the comment regarding P>.05 referred to, but have removed spaces and made it non-italics.

Reviewer #2 (Recommendations for the authors):

Statistical analyses and their outcomes are clearly indicated only for the data in Figure 1 and Figure 5 and in the supplement for Figure 1, while they are not reported/not easily accessible for other data. For the main figures, statistics should be indicated in the figure for an easier assessment of the data. In case of multiple comparisons potentially crowding the plots too much, statistics may be in a supplementary file/table.

See response above.

In case of the hemocytes, besides phagocytosis, I would think that ROS generation via the DUOX/NOX system is also an integral part of the immune response against pathogens, and that has not been included here. That might be an interesting addition for future experiments. As the NimC1, eater double mutant flies are said to have fewer hemocytes, it is possible that this function of the hemocytes is affected as well. This could be commented on in the text.

The reviewer raises a good point. The role of DUOX and NOX in ROS responses is not assessed in our study. To our knowledge, DUOX and NOX participate primarily in the wound repair response, or in epithelial renewal at damage sites or in the gut. In our study on systemic immunity, we did not assess the role of clotting, the precise function of ROS, and we have missed other host defense or stress response mechanisms as well (e.g. constitutively-expressed AMP-like genes, TEPs, JAK-STAT) that likely play a role in the systemic immune defense. Considering the lethality caused by Nox and Duox mutation, there would be inherent genetic difficulties to recombine these as multiple mutations. Unfortunately, this makes it difficult to include these processes in our analysis in a systematic manner. We are already happy to have generated fly lines lacking four immune modules simultaneously, even if they are not fully immune deficient. We have mentioned this point in the discussion (Line 613-on).

Of note, the NimC1, eater double mutants actually have decreased hemocyte counts at the adult stage (Melcarne et al., 2019). Thus NimC1, eater double mutants are not impaired only in phagocytosis, but the overall cellular response. We make a point to outline this in Line 225-257, and 607.

I think it could be mentioned that the melanization response at larval stage (against parasitoids) functions differently from the melanization described here (requiring hemocyte differentiation and PPO3).

A good point. We have added this mention in Line 97:

“In addition, a third PPO gene (PPO3) is specifically expressed by lamellocytes, specialized hemocytes that differentiate in larvae responding to and enveloping invading parasites (Dudzik et al., 2015)”.

Overall, the clarity of the figures and figure legends could be worked on to make them a bit easier to follow. Below are some of my suggestions:

(1) In Figure 2, adding headings to parts C & D (similarly to A & B) would make it easier to follow what is happening in the figure at a glance. Also, it is rather difficult to visually follow which strain is which in the plots. I'd suggest adding the key/legend for single mutants below 2A & B, and the key for the double mutants below C & D. If a mutant is present in A & B and in C & D, it could be included in both keys. I also think that it would be intuitive to present the single mutants by dashed lines and double mutants by continuous lines (or vice versa), so that one would easily distinguish between them. Of note, the figure legend says that A & B are single mutants, but for example in B there are also some double mutants (?).

We have modified Figure 2 and Figure 2supp1 labelling. We also realise there was an error in the column titling that contributed to the confusion. We hope the new layout is clear, and thank the reviewers for noting this issue.

(2) In Figure 3, it looks like ΔMel is almost identical to controls in the clean injury survival, but in Figure 2C, it is clearly doing worse. I might be missing something here, but would

like the authors to clarify the matter. Also, the meaning of the numbers in the heat map could be explained in the figure legend and/or added to the figure (color key).

The reviewer is correct. We thank the reviewer for this astute observation. Inadvertently, we used an old version of the Figure 2 preparation where only a subset of experiments was entered in the Prism data file rather than the total data used to inform Figure 3. This issue affected all genotypes.

We have reviewed the data in Figure 2, Figure 2supp1, and Figure 3, and updated these figures accordingly to ensure they represent the full survival data. We have also incorporated new experiments into the sum data related to male-female differences and to fill gaps in the data from the 1st submission. We will also note due to the nature of 1st decimal rounding that the difference between WT and Δ MeI appears slightly underrepresented: the true difference (over the 7-day lifespan) is 0.37. We've provided a version of this figure rounded to 2 decimal places below, but prefer the simpler 1 decimal place in the main text for readability. The updated Figure 2 shows the full data in Figure 3 accurately.

We will also take this opportunity to highlight how conservative our ≥ 1.0 days difference approach is. Breaking down survival curve patterns in Figure 2 relative to mean differences in Figure 3, for clean injury, approximately ~75% of Δ MeI flies survive to day 7 with mortality mostly taking place between days 3-7. The result is a mean lifespan of 6.37 days. On a survival curve, this difference appears quite strong, but in our mean lifespan table the difference is rather muted (WT vs. Δ MeI difference = 0.37 days). Thus, differences of ≥ 1.0 days reflect very strong trends in survival data that are near-guaranteed to be independent of experimental noise. While we note issues that prevented us from a fully systematic sampling for all experiments, we are confident that the ≥ 1.0 day differences we highlight, using the rules explained in the main text, are robust. While this approach could be seen as overly conservative, it is our preference in this initial study, containing combinations of 25 treatments and 14 genotypes, to be highly conservative. Future studies may investigate other strong differences we have not highlighted, and the data we provide here can help generate expectations and guide those studies.

Author response image 1.

Figure 3 with 2 decimals places of rounding for mean lifespans. The 7-day clean injury mean lifespan of WT is 6.74 days, and of Δ MeI is 6.37 days. Due to rounding, in the 1 decimal Figure 3 this difference appears as if it is only 0.3 days, but it closer to 0.4 days. Regardless, this level of difference, which appears rather clearly in a survival curve, is well below the level of difference we have chosen to highlight in our study.

Type	Species	Temp	Dose	Days	N exp	WT	Δ AMP4	Δ Bom	Δ MD	Δ TOLL	Δ Phag	Δ MeI	Δ Phag, Δ MeI	Δ TOLL, Δ MeI ^{WT}	Δ TOLL, Δ MeI ^{MD}	Δ MD, Δ MeI	Δ MD, Δ Phag	Δ MD, Δ TOLL	Δ TPM
Controls	Unchallenged	25°C	in	7	13	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90	6.90
	Clean injury	25°C	in	7	36	6.74	6.55	6.52	6.74	6.65	6.81	6.37	5.28	4.62	5.37	4.70	6.53	6.23	5.50
	Clean injury	25°C	in	7	11	6.59	6.45	5.92	6.28	6.08	6.55	6.36	4.90	2.97	4.82	3.68	5.98	5.60	3.89
	Clean injury	25°C	in	10	36	6.56	6.13	5.91	6.24	6.34	6.59	6.45	6.18	5.23	6.26	5.56	6.13	6.87	6.34
	Unchallenged	25°C	in	10	11	9.94	9.27	9.52	9.52	7.12	9.28	9.22	8.73	5.94	9.30	9.45	9.12	9.45	7.67
Viruses	Drosophila C virus	25°C	1000 TCID ₅₀	10	5	8.28	6.71	7.71	6.82	7.85	8.13	6.29	6.33	5.46	5.06	5.14	6.68	3.80	6.13
	Flock House virus	25°C	20000 TCID ₅₀	10	5	8.55	7.14	7.19	7.28	7.26	7.32	6.87	6.18	5.05	6.05	4.92	6.79	5.20	5.06
	Drosophila X virus	25°C	431000 TCID ₅₀	10	2	9.03	6.86	8.16	7.51	7.90	7.80	5.37	7.44	4.91	5.08	6.00	5.17	5.63	4.63
	Arbovirus reovirus-6 virus	25°C	10100 TCID ₅₀	10	3	9.38	7.47	9.36	8.97	8.36	8.78	5.64	6.63	5.78	5.82	6.13	6.28	5.85	4.41
	Sindbis virus	25°C	50000 TCID ₅₀	10	3	9.37	8.44	8.63	8.91	8.79	8.28	5.42	5.90	4.93	4.86	6.01	6.96	4.51	4.33
Fungi	Aspergillus fumigatus (natural)	25°C	inlet on plate	10	5	10.00	9.20	9.90	8.76	7.24	7.57	9.05	8.24	6.76	8.57	7.87	8.73	6.88	4.67
	Beauveria bassiana (natural)	25°C	50mg	10	5	9.63	8.22	9.75	9.45	7.27	7.79	8.73	7.73	5.03	6.00	6.88	6.39	5.60	4.39
	Beauveria bassiana	25°C	OD=1	7	6	6.96	5.96	3.96	5.79	3.85	5.21	5.30	4.43	5.03	3.40	4.38	4.67	3.14	3.27
	Candida albicans	25°C	OD=200	7	8	6.66	5.10	5.59	5.56	3.72	6.60	5.87	6.44	1.75	2.79	5.72	5.88	3.22	3.38
	Beauveria bassiana	25°C	OD=1	7	6	6.11	6.04	6.51	6.50	5.65	5.17	5.83	5.68	4.06	5.75	5.12	5.06	6.32	3.89
Gram positive	Microbacterium maris	25°C	OD=100	7	3	5.96	6.23	6.49	5.58	5.11	6.62	4.60	5.84	3.25	4.38	4.86	5.92	4.52	2.96
	Corynebacterium jeikeium	25°C	OD=200	7	8	6.71	5.68	6.83	6.24	6.40	6.48	6.46	5.64	4.57	6.46	4.82	6.31	2.06	2.05
	Streptococcus pneumoniae	25°C	OD=0.3	7	3	6.78	6.10	6.40	6.69	6.61	6.32	6.22	6.37	5.68	5.03	5.13	6.02	5.45	4.52
	Enterococcus faecalis	25°C	OD=0.5	7	8	6.62	4.54	5.95	4.24	5.05	5.23	5.11	3.88	1.69	4.19	4.03	3.98	1.27	1.27
	Staphylococcus aureus	25°C	OD=0.5	7	8	5.05	4.83	5.67	4.24	3.44	3.63	2.37	1.88	1.96	2.04	2.13	4.08	3.43	1.85
	Listeria monocytogenes	25°C	OD=0.5	7	5	4.84	4.74	2.57	2.54	1.94	4.56	5.25	3.95	2.06	2.19	3.50	3.26	1.97	1.58
	Bacillus subtilis	25°C	OD=1	7	7	5.71	3.53	4.05	3.39	3.82	5.96	5.41	4.51	2.81	2.71	4.49	5.40	2.15	3.02
	Microbacterium maris	25°C	OD=10	7	2	3.10	1.10	2.90	1.10	1.34	2.84	2.27	3.23	1.78	1.93	1.90	1.13	1.50	1.50
	Providencia rettgeri	25°C	OD=0.1	7	7	5.47	1.11	5.09	1.96	4.41	3.64	1.84	1.48	1.51	5.30	1.27	2.10	1.90	1.41
	Providencia burhodogyanensis	25°C	OD=10	7	7	5.09	1.72	4.58	1.62	4.63	4.90	3.98	4.24	2.92	4.90	1.51	1.90	1.34	1.29
Gram negative	Providencia carolinensis	25°C	OD=1	7	8	6.42	1.46	5.75	1.82	5.76	4.43	5.79	5.29	4.97	5.22	1.82	1.75	1.42	1.44
	Klebsiella pneumoniae	25°C	OD=1	7	4	6.70	1.40	6.65	1.40	6.80	6.93	5.82	6.33	3.93	6.93	1.32	2.00	1.90	1.33
	Enterobacter cloacae	25°C	OD=10	7	8	6.59	3.65	6.73	3.65	6.73	5.96	6.40	6.13	4.92	5.90	3.32	3.98	1.61	1.32
	Enterobacter coli	25°C	OD=200	7	8	6.57	5.04	6.90	3.83	6.19	5.40	5.47	5.51	3.84	5.25	3.80	4.62	1.71	1.62
	Salmonella enterica ser. typhimurium	25°C	OD=200	7	4	6.64	4.78	6.67	3.51	6.30	6.40	6.33	6.56	5.45	4.84	4.72	3.89	2.68	1.84

(1) Figure 4: I find it very tedious to compare CFUs among different mutants from the plots. As the idea is to compare bacterial loads among the mutants at different

timepoints, it would be easier to compare them if the data were shown within a timepoint (CFUs of each mutant at 2h, at 6h, and so on). This is also how the results are written in the text (within a time point). Would it also be clearer if the CFU plots were named, for example: "A", "B", and "C"?

We appreciate this note. We feel both representations have merits and pitfalls, but prefer our original design showing the progression of bacterial growth within genotype first. However, we have added dotted lines representing the wild-type bacterial loads at 2hpi, 12hpi, and 24hpi to assist the reader in making acrossgenotype comparisons at key time points. Like this, the reader can see if the error bars (StDev) overlap the mean of the wild-type, and so make more intuitive judgements about whether these differences are meaningful.

(2) Figure 2D is not referred to in the text.

We now cite Fig2D in Line 356.

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